

HOMEWORK 1 (110 points + 1) : CAUTION! CONTENTS ARE HOT 🌋

****DUE: *SEPTEMBER 18, 2025 @ 11:59 PM*****

****24-HR LATE DUE DATE WITH A 15% PENALTY:
*SEPTEMBER 19, 2025 @ 11:59 PM*****

The [NCEI/WDS Global Significant Volcanic Eruptions Database](#) is a very comprehensive collection of +600 volcanic eruptions dating from 4360 BC to the present. Due to the nature of this assignment, we will be dealing with relatively newer volcanoes (in which some are still obviously still older than anyone on Earth currently). Each eruption in the database is classified as significant if it meets one or more criteria, such as causing fatalities, incurring **damage on property (+\$1 million)**, reaching a **Volcanic Explosivity Index (VEI) of 6 or higher**, generating a tsunami, or being linked to a significant earthquake. The VEI is a scale that measures the explosiveness of volcanic eruptions, providing insight into the magnitude and potential consequences of the eruptions. The database includes detailed information on the location, type of volcano, last known eruption, VEI, casualties, property damage, and much more.



We are going to dive straight into these volcanoes (well... their dataset), to swim our way into Pandas proficiency!

You will find the [Pandas Documentation](#) helpful. There are also some helpful links to guide you along the way! Don't get burned 🔥

For this assignment, we are including a special bonus question to help everyone get familiar with the proper assignment submission routine. We strongly encourage you to complete Bonus

Question 1, learn the process, and follow this practice for all future assignments.

For this assignment only, you will not lose points if you skip the bonus question. However, starting from Homework 2, submitting both the Jupyter Notebook file and the PDF will be required in order to receive full credit.

DO NOT REMOVE ANY PART OF ANY OF THE QUESTIONS OR YOU LOSE CREDIT

No Hardcoding either 😊❤️

REMEMBER TO SHOW OUTPUT

Part 1: Maintenance 🌟 (25 POINTS TOTAL)

First, we're going to familiarize ourselves with the process. As in most languages, Python looks best when its modules are imported first before any other code is written ✨

```
In [29]: # Make sure these code blocks run properly and that you have properly installed the
import pandas as pd
import requests
# import other libraries here

# Don't Remove this
pd.set_option('display.max_rows', None)
pd.set_option('display.max_columns', None)
```

As you may have noticed, there's another library aside from Pandas called "[requests](#)." **The requests library allows you to send HTTP requests to a server, retrieve the content, and process it at ease.** It's very beginner friendly for those attempting to get into webscraping (super important for collecting and creating datasets). We also recommend looking into [BeautifulSoup](#) (yeah, soup LOL), another wonderful library that can be paired with the requests library for webscraping.

As shown below, sometimes specific websites require specific headers in order to process a request to access the data.

To check if a request was processed successfully, use the [status_code](#) function to see if the process returned 200.

```
In [30]: # API URL and headers in case request gets denied.
api_url = "https://www.ngdc.noaa.gov/hazel/hazard-service/api/v1/volcanoes"

headers = {
    'accept': '*/*'
}
```

TASK 1.0: Cute Webscraping (5 points)

To make our cute webscraper we need to **create a GET request** using the relevant information given above.

This particular dataset NOAA returns data from the API as **json** when a user makes a request. The json data has a particular format, so we will extract our needed information only from the field called **items** to make a dataframe.

After properly scraping the data, name this dataframe *df*

Save this dataframe into a CSV file named 'volcanoes.csv'

You won't need to run this more than once

```
In [31]: response = requests.get(api_url, headers=headers)

if response.status_code == 200:
    data = response.json()
    items = data['items']
    df = pd.DataFrame(items)
    df.to_csv('volcanoes.csv', index=False)
```

TASK 1.1: 1 Liner Thingz (3 points)

We need to get an idea of what this dataset is going to look. In order to do that, let's take a look at some of the most **basic things** our dataframe has.

Read the directions carefully and code your answer with only one line of code.

CAN'T USE LOOPS. DO NOT DISPLAY THE DATAFRAME, JUST YOUR CODE OUTPUT HERE.

1.1.1: In one line of code and **using only one single method call**, output the **summary statistics** (count, mean, std, min, max, quartiles) of all numerical features in the dataframe.

Hint: Look for a method that gives a descriptive overview of numerical columns.

```
In [32]: df.describe()
```

Out[32]:

	id	year	month	day	tsunamiEventId	earthquakeEventId
count	200.000000	200.000000	177.000000	155.000000	37.000000	17.000000
mean	375.635000	1809.800000	6.293785	15.825806	1818.648649	3632.117647
std	306.114755	248.034282	3.469611	9.185073	1742.563313	3015.859162
min	1.000000	-141.000000	1.000000	1.000000	61.000000	421.000000
25%	78.500000	1774.500000	3.000000	8.000000	613.000000	1328.000000
50%	348.000000	1880.500000	7.000000	15.000000	1181.000000	2179.000000
75%	664.250000	1950.000000	9.000000	24.000000	1977.000000	6195.000000
max	994.000000	1996.000000	12.000000	31.000000	5863.000000	10519.000000



1.1.2: In one line of code and **using only one single method call**, print the **summary information** of the dataframe, including index dtype, column names, non-null counts, and memory usage.

In [33]:

```
df.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 43 columns):
 #   Column                                     Non-Null Count  Dtype
---  -
 0   id                                         200 non-null    int64
 1   year                                       200 non-null    int64
 2   month                                     177 non-null    float64
 3   day                                        155 non-null    float64
 4   tsunamiEventId                           37 non-null     float64
 5   earthquakeEventId                        17 non-null     float64
 6   volcanoLocationId                        200 non-null    int64
 7   volcanoLocationNewNum                    200 non-null    int64
 8   volcanoLocationNum                       200 non-null    object
 9   name                                      200 non-null    object
10   location                                  200 non-null    object
11   country                                   200 non-null    object
12   latitude                                  200 non-null    float64
13   longitude                                 200 non-null    float64
14   elevation                                200 non-null    int64
15   morphology                               200 non-null    object
16   agent                                     128 non-null    object
17   deathsTotal                             128 non-null    float64
18   deathsAmountOrderTotal                   180 non-null    float64
19   damageAmountOrderTotal                   59 non-null     float64
20   significant                              200 non-null    bool
21   publish                                  200 non-null    bool
22   eruption                                 200 non-null    bool
23   status                                   200 non-null    object
24   timeErupt                               200 non-null    object
25   vei                                       192 non-null    float64
26   deathsAmountOrder                       169 non-null    float64
27   damageAmountOrder                       51 non-null     float64
28   housesDestroyedAmountOrderTotal          52 non-null     float64
29   deaths                                   118 non-null    float64
30   injuries                                 30 non-null     float64
31   injuriesAmountOrder                     39 non-null     float64
32   injuriesTotal                           33 non-null     float64
33   injuriesAmountOrderTotal                 41 non-null     float64
34   housesDestroyedAmountOrder               42 non-null     float64
35   housesDestroyed                         13 non-null     float64
36   housesDestroyedTotal                     16 non-null     float64
37   missingAmountOrder                       3 non-null     float64
38   missingAmountOrderTotal                  3 non-null     float64
39   missing                                  1 non-null     float64
40   missingTotal                             1 non-null     float64
41   damageMillionsDollars                    1 non-null     float64
42   damageMillionsDollarsTotal               1 non-null     float64
dtypes: bool(3), float64(27), int64(5), object(8)
memory usage: 63.2+ KB

```

We won't be using some of the data because there is a lot of missing data.

1.1.3: In one line of code, create a **new dataframe** called **new_df** that **discards** all the features of the **old** dataframe **except for the following**:

```
id, year, month, day, tsunamiEventId, earthquakeEventId,
volcanoLocationId, volcanoLocationNewNum, name, country, elevation,
morphology, deathsTotal, vei, deaths
```

Hint: Don't use any drop function

```
In [34]: new_df = df[["id", "year", "month", "day", "tsunamiEventId", "earthquakeEventId", "v
```

TASK 1.2: 1 Liner Shenaniganz (7 points)

We're going to tidy up the **new dataframe** a little more with some more advanced 1 liner code.

Read the directions carefully and code your answer with only one line of code.

For this section, keep the method of display that is already in the box. Write your code as indicated.

YOU CAN'T USE ONE LINE LOOPS OR ANY KIND OF LOOP.

1.2.1: *In one line of code, **drop any row** that contains **NaN** in **any one** of the columns indicating a measure of **time**.*

```
In [35]: new_df = new_df.dropna(subset=['year', 'month', 'day'])

print(new_df)
```

	id	year	month	day	tsunamiEventId	earthquakeEventId	\
0	1	1169	2.0	4.0	2852.0	421.0	
1	2	1329	7.0	15.0	NaN	NaN	
3	4	1888	3.0	13.0	1175.0	NaN	
5	6	1832	11.0	1.0	NaN	NaN	
6	7	1977	1.0	10.0	NaN	NaN	
8	9	1779	8.0	8.0	NaN	NaN	
10	11	1907	8.0	4.0	NaN	NaN	
11	12	1905	3.0	10.0	NaN	NaN	
12	13	1986	7.0	24.0	NaN	NaN	
13	14	1536	3.0	26.0	NaN	NaN	
14	15	1904	2.0	25.0	NaN	NaN	
15	16	1878	2.0	4.0	1095.0	6195.0	
16	17	1737	5.0	20.0	NaN	NaN	
17	18	1917	4.0	1.0	NaN	NaN	
19	20	1979	3.0	8.0	NaN	NaN	
20	21	1970	3.0	2.0	NaN	NaN	
22	23	1682	8.0	12.0	NaN	NaN	
23	24	1805	8.0	11.0	NaN	NaN	
24	25	1919	5.0	22.0	1476.0	NaN	
25	26	1930	9.0	11.0	4218.0	NaN	
26	27	1903	8.0	30.0	NaN	NaN	
27	28	1928	11.0	8.0	NaN	NaN	
28	29	1872	4.0	24.0	NaN	NaN	
29	30	1979	9.0	12.0	NaN	NaN	
30	31	1987	4.0	17.0	NaN	NaN	
31	32	1914	9.0	10.0	NaN	NaN	
32	33	1886	8.0	31.0	NaN	NaN	
33	34	1912	12.0	3.0	NaN	NaN	
34	35	1895	6.0	17.0	NaN	NaN	
35	36	1538	9.0	29.0	NaN	NaN	
36	37	1853	6.0	24.0	NaN	NaN	
37	38	1954	8.0	3.0	NaN	NaN	
38	39	1944	3.0	27.0	NaN	NaN	
39	40	1984	8.0	15.0	5807.0	NaN	
41	42	1926	1.0	23.0	NaN	NaN	
42	43	1631	2.0	14.0	NaN	920.0	
43	44	1843	11.0	25.0	NaN	NaN	
44	45	1944	12.0	4.0	NaN	NaN	
45	49	1979	2.0	20.0	NaN	NaN	
47	64	1672	8.0	4.0	NaN	NaN	
48	69	1822	12.0	27.0	NaN	NaN	
49	71	1832	12.0	25.0	NaN	NaN	
50	81	1872	4.0	15.0	NaN	NaN	
51	82	1872	11.0	3.0	NaN	NaN	
53	92	1920	7.0	25.0	NaN	NaN	
54	94	1930	12.0	18.0	NaN	NaN	
55	100	1954	1.0	18.0	NaN	NaN	
56	104	1961	5.0	8.0	NaN	NaN	
57	105	1969	1.0	7.0	NaN	NaN	
58	111	1986	12.0	31.0	NaN	NaN	
59	112	1994	11.0	22.0	NaN	NaN	
65	128	1716	7.0	20.0	NaN	NaN	
66	132	1826	10.0	13.0	NaN	NaN	
67	133	1848	5.0	16.0	NaN	NaN	
68	135	1864	1.0	4.0	NaN	NaN	

69	136	1875	1.0	29.0	NaN	NaN
70	137	1901	5.0	22.0	NaN	NaN
71	138	1919	5.0	19.0	NaN	NaN
72	140	1951	8.0	31.0	NaN	NaN
73	141	1966	4.0	26.0	NaN	NaN
74	144	1990	2.0	10.0	NaN	NaN
76	167	1885	4.0	18.0	NaN	NaN
77	176	1895	5.0	22.0	NaN	NaN
78	191	1909	8.0	29.0	NaN	NaN
79	193	1911	11.0	8.0	NaN	NaN
80	195	1913	6.0	23.0	NaN	NaN
81	196	1941	9.0	21.0	NaN	NaN
83	199	1946	10.0	29.0	NaN	NaN
84	200	1950	8.0	28.0	NaN	NaN
85	201	1963	5.0	5.0	NaN	NaN
86	202	1967	8.0	31.0	NaN	NaN
87	203	1985	5.0	10.0	NaN	NaN
88	204	1978	9.0	19.0	NaN	NaN
89	205	1976	8.0	31.0	NaN	NaN
90	206	1981	3.0	29.0	NaN	NaN
98	333	1817	1.0	24.0	NaN	NaN
99	345	1963	9.0	5.0	NaN	NaN
100	351	1963	3.0	18.0	1942.0	NaN
101	353	1963	5.0	16.0	NaN	4292.0
102	367	1815	4.0	10.0	613.0	NaN
103	389	1969	1.0	28.0	NaN	NaN
104	394	1928	8.0	4.0	1607.0	NaN
105	395	1964	1.0	1.0	NaN	NaN
107	398	1981	9.0	5.0	NaN	NaN
108	410	1869	7.0	7.0	NaN	NaN
109	411	1907	9.0	28.0	NaN	NaN
111	441	1948	4.0	7.0	NaN	NaN
113	447	1983	8.0	17.0	4702.0	NaN
115	457	1659	11.0	11.0	285.0	NaN
117	466	1692	6.0	4.0	NaN	NaN
120	477	1694	11.0	20.0	NaN	NaN
121	482	1820	6.0	11.0	NaN	NaN
122	486	1988	5.0	9.0	NaN	NaN
123	488	1983	7.0	23.0	NaN	NaN
124	489	1845	2.0	8.0	741.0	1865.0
125	525	1958	7.0	12.0	NaN	NaN
127	533	1871	3.0	3.0	1022.0	NaN
128	552	1940	6.0	20.0	NaN	NaN
129	561	1974	2.0	11.0	NaN	NaN
130	562	1976	9.0	15.0	NaN	NaN
131	569	1889	9.0	6.0	1181.0	6309.0
132	578	1711	12.0	11.0	NaN	NaN
133	580	1812	8.0	6.0	NaN	NaN
134	581	1856	3.0	2.0	859.0	1995.0
135	584	1892	6.0	7.0	1197.0	NaN
136	585	1913	3.0	14.0	1407.0	2973.0
137	589	1966	8.0	12.0	NaN	NaN
139	599	1673	5.0	20.0	312.0	NaN
140	607	1608	7.0	18.0	222.0	NaN
141	617	1772	5.0	9.0	504.0	NaN
142	618	1773	2.0	2.0	NaN	NaN

144	627	1838	2.0	26.0	NaN	NaN
145	629	1840	2.0	2.0	718.0	NaN
146	643	1871	8.0	7.0	NaN	NaN
147	657	1962	12.0	31.0	NaN	NaN
149	664	1646	7.0	19.0	5754.0	10519.0
151	666	1861	12.0	29.0	NaN	NaN
152	669	1890	6.0	29.0	NaN	NaN
153	677	1873	1.0	16.0	NaN	NaN
154	679	1871	4.0	30.0	1020.0	2179.0
155	680	1950	9.0	15.0	NaN	NaN
156	693	1996	8.0	10.0	NaN	NaN
157	697	1933	12.0	25.0	1666.0	NaN
158	698	1978	7.0	29.0	NaN	NaN
159	704	1766	7.0	20.0	NaN	NaN
160	706	1800	10.0	30.0	NaN	NaN
161	707	1814	2.0	1.0	NaN	NaN
162	712	1853	7.0	13.0	NaN	NaN
164	716	1871	12.0	8.0	NaN	NaN
165	724	1887	3.0	9.0	NaN	NaN
166	730	1897	5.0	23.0	NaN	NaN
168	733	1938	6.0	5.0	NaN	NaN
169	735	1947	1.0	7.0	NaN	NaN
170	736	1968	4.0	21.0	NaN	NaN
171	739	1984	9.0	9.0	NaN	NaN
173	752	1716	9.0	24.0	387.0	1205.0
174	753	1749	8.0	11.0	431.0	1304.0
175	754	1754	11.0	28.0	2853.0	1328.0
176	756	1874	7.0	19.0	NaN	NaN
177	761	1911	1.0	30.0	1392.0	8391.0
178	762	1965	9.0	28.0	1977.0	NaN
179	777	1969	3.0	21.0	2872.0	NaN
180	787	1853	10.0	29.0	809.0	NaN
182	823	1933	12.0	24.0	NaN	NaN
184	844	766	7.0	20.0	61.0	7341.0
185	846	1471	11.0	3.0	NaN	NaN
186	858	1779	11.0	8.0	NaN	NaN
187	859	1781	4.0	11.0	529.0	NaN
188	867	1914	1.0	12.0	1412.0	2994.0
190	878	1955	10.0	13.0	NaN	NaN
191	904	1566	10.0	31.0	NaN	NaN
192	915	1716	11.0	9.0	NaN	NaN
193	916	1717	2.0	7.0	NaN	NaN
194	932	1792	5.0	21.0	568.0	1515.0
195	933	1991	6.0	3.0	NaN	NaN

	volcanoLocationId	volcanoLocationNewNum	name \
0	10106	211060	Etna
1	10106	211060	Etna
3	50107	251070	Ritter Island
5	10106	211060	Etna
6	20303	223030	Nyiragongo
8	10102	211020	Vesuvius
10	201112	221112	Alayta
11	10102	211020	Vesuvius
12	10104	211040	Stromboli
13	10106	211060	Etna

14	30301	233010	Karthala
15	50214	252140	Rabaul
16	10102	211020	Vesuvius
17	40105	241050	Okataina
19	50103	251030	Karkar
20	10101	211010	Campi Flegrei
22	10102	211020	Vesuvius
23	10102	211020	Vesuvius
24	10104	211040	Stromboli
25	10104	211040	Stromboli
26	40105	241050	Okataina
27	10106	211060	Etna
28	10102	211020	Vesuvius
29	10106	211060	Etna
30	10106	211060	Etna
31	40104	241040	Whakaari/White Island
32	40311	243110	Niuafu'ou
33	20302	223020	Nyamulagira
34	50103	251030	Karkar
35	10101	211010	Campi Flegrei
36	40311	243110	Niuafu'ou
37	50101	251010	Bam
38	10102	211020	Vesuvius
39	20403	224030	Oku Volcanic Field
41	10204	212040	Santorini
42	201141	221141	Dama Ali
43	10106	211060	Etna
44	60320	263200	Dieng Volcanic Complex
45	60320	263200	Dieng Volcanic Complex
47	60325	263250	Merapi
48	60325	263250	Merapi
49	60325	263250	Merapi
50	60325	263250	Merapi
51	60325	263250	Merapi
53	60325	263250	Merapi
54	60325	263250	Merapi
55	60325	263250	Merapi
56	60325	263250	Merapi
57	60325	263250	Merapi
58	60325	263250	Merapi
59	60325	263250	Merapi
65	60328	263280	Kelud
66	60328	263280	Kelud
67	60328	263280	Kelud
68	60328	263280	Kelud
69	60328	263280	Kelud
70	60328	263280	Kelud
71	60328	263280	Kelud
72	60328	263280	Kelud
73	60328	263280	Kelud
74	60328	263280	Kelud
76	60330	263300	Semeru
77	60330	263300	Semeru
78	60330	263300	Semeru
79	60330	263300	Semeru
80	60330	263300	Semeru

81	60330	263300	Semeru
83	60330	263300	Semeru
84	60330	263300	Semeru
85	60330	263300	Semeru
86	60330	263300	Semeru
87	60330	263300	Semeru
88	60330	263300	Semeru
89	60330	263300	Semeru
90	60330	263300	Semeru
98	60335	263350	Ijen
99	60401	264010	Batur
100	60402	264020	Agung
101	60402	264020	Agung
102	60404	264040	Tambora
103	60411	264110	Iya
104	60415	264150	Paluweh
105	60415	264150	Paluweh
107	60415	264150	Paluweh
108	60418	264180	Lewotobi
109	60418	264180	Lewotobi
111	60425	264250	Iliwerung
113	60425	264250	Iliwerung
115	60505	265050	Teon
117	60507	265070	Serua
120	60509	265090	Banda Api
121	60509	265090	Banda Api
122	60509	265090	Banda Api
123	60601	266010	Colo
124	60603	266030	Soputan
125	60611	266110	Mahawu
127	60701	267010	Ruang
128	60702	267020	Karangetang
129	60702	267020	Karangetang
130	60702	267020	Karangetang
131	60703	267030	Banua Wuhu
132	60704	267040	Awu
133	60704	267040	Awu
134	60704	267040	Awu
135	60704	267040	Awu
136	60704	267040	Awu
137	60704	267040	Awu
139	60804	268040	Gamkonora
140	60806	268060	Gamalama
141	60806	268060	Gamalama
142	60806	268060	Gamalama
144	60806	268060	Gamalama
145	60806	268060	Gamalama
146	60806	268060	Gamalama
147	60806	268060	Gamalama
149	60807	268070	Kie Besi
151	60807	268070	Kie Besi
152	60807	268070	Kie Besi
153	70106	271060	Ragang
154	70108	271080	Camiguin
155	70108	271080	Camiguin
156	70202	272020	Kanlaon

157	70301	273010	Bulusan
158	70301	273010	Bulusan
159	70303	273030	Mayon
160	70303	273030	Mayon
161	70303	273030	Mayon
162	70303	273030	Mayon
164	70303	273030	Mayon
165	70303	273030	Mayon
166	70303	273030	Mayon
168	70303	273030	Mayon
169	70303	273030	Mayon
170	70303	273030	Mayon
171	70303	273030	Mayon
173	70307	273070	Taal
174	70307	273070	Taal
175	70307	273070	Taal
176	70307	273070	Taal
177	70307	273070	Taal
178	70307	273070	Taal
179	70402	274020	Didicas
180	80103	281030	Unnamed
182	80205	282050	Kuchinoerabujima
184	80208	282080	Aira
185	80208	282080	Aira
186	80208	282080	Aira
187	80208	282080	Aira
188	80208	282080	Aira
190	80208	282080	Aira
191	80209	282090	Kirishimayama
192	80209	282090	Kirishimayama
193	80209	282090	Kirishimayama
194	80210	282100	Unzendake
195	80210	282100	Unzendake

	country	elevation	morphology	deathsTotal	vei \
0	Italy	3357	Stratovolcano	16000.0	NaN
1	Italy	3357	Stratovolcano	NaN	3.0
3	Papua New Guinea	75	Stratovolcano	NaN	2.0
5	Italy	3357	Stratovolcano	NaN	2.0
6	Congo, DRC	3470	Stratovolcano	NaN	1.0
8	Italy	1281	Complex volcano	NaN	3.0
10	Ethiopia	1496	Shield volcano	NaN	2.0
11	Italy	1281	Complex volcano	1.0	2.0
12	Italy	924	Stratovolcano	1.0	2.0
13	Italy	3357	Stratovolcano	1.0	3.0
14	Comoros	2361	Shield volcano	1.0	2.0
15	Papua New Guinea	688	Pyroclastic shield	1.0	3.0
16	Italy	1281	Complex volcano	2.0	3.0
17	New Zealand	1111	Lava dome	2.0	1.0
19	Papua New Guinea	1839	Stratovolcano	2.0	2.0
20	Italy	458	Caldera	3.0	NaN
22	Italy	1281	Complex volcano	4.0	3.0
23	Italy	1281	Complex volcano	4.0	2.0
24	Italy	924	Stratovolcano	4.0	3.0
25	Italy	924	Stratovolcano	6.0	3.0
26	New Zealand	1111	Lava dome	4.0	1.0

27	Italy	3357	Stratovolcano	5.0	1.0
28	Italy	1281	Complex volcano	9.0	3.0
29	Italy	3357	Stratovolcano	9.0	3.0
30	Italy	3357	Stratovolcano	2.0	2.0
31	New Zealand	294	Stratovolcano	11.0	NaN
32	Tonga	260	Shield volcano	NaN	4.0
33	Congo, DRC	3058	Shield volcano	20.0	3.0
34	Papua New Guinea	1839	Stratovolcano	21.0	2.0
35	Italy	458	Caldera	24.0	3.0
36	Tonga	260	Shield volcano	25.0	0.0
37	Papua New Guinea	685	Stratovolcano	25.0	2.0
38	Italy	1281	Complex volcano	27.0	3.0
39	Cameroon	3011	Maar	37.0	NaN
41	Greece	367	Shield volcano	48.0	2.0
42	Ethiopia	1068	Shield volcano	50.0	NaN
43	Italy	3357	Stratovolcano	59.0	2.0
44	Indonesia	2565	Complex volcano	117.0	2.0
45	Indonesia	2565	Complex volcano	149.0	1.0
47	Indonesia	2910	Stratovolcano	3000.0	3.0
48	Indonesia	2910	Stratovolcano	100.0	3.0
49	Indonesia	2910	Stratovolcano	32.0	2.0
50	Indonesia	2910	Stratovolcano	30.0	4.0
51	Indonesia	2910	Stratovolcano	200.0	4.0
53	Indonesia	2910	Stratovolcano	35.0	2.0
54	Indonesia	2910	Stratovolcano	1369.0	3.0
55	Indonesia	2910	Stratovolcano	64.0	2.0
56	Indonesia	2910	Stratovolcano	6.0	3.0
57	Indonesia	2910	Stratovolcano	3.0	2.0
58	Indonesia	2910	Stratovolcano	1.0	2.0
59	Indonesia	2910	Stratovolcano	66.0	3.0
65	Indonesia	1730	Stratovolcano	NaN	2.0
66	Indonesia	1730	Stratovolcano	NaN	4.0
67	Indonesia	1730	Stratovolcano	22.0	3.0
68	Indonesia	1730	Stratovolcano	54.0	2.0
69	Indonesia	1730	Stratovolcano	NaN	NaN
70	Indonesia	1730	Stratovolcano	NaN	3.0
71	Indonesia	1730	Stratovolcano	5110.0	4.0
72	Indonesia	1730	Stratovolcano	7.0	4.0
73	Indonesia	1730	Stratovolcano	215.0	4.0
74	Indonesia	1730	Stratovolcano	35.0	4.0
76	Indonesia	3657	Stratovolcano	74.0	2.0
77	Indonesia	3657	Stratovolcano	80.0	2.0
78	Indonesia	3657	Stratovolcano	221.0	2.0
79	Indonesia	3657	Stratovolcano	NaN	3.0
80	Indonesia	3657	Stratovolcano	NaN	2.0
81	Indonesia	3657	Stratovolcano	NaN	2.0
83	Indonesia	3657	Stratovolcano	NaN	2.0
84	Indonesia	3657	Stratovolcano	6.0	1.0
85	Indonesia	3657	Stratovolcano	1.0	2.0
86	Indonesia	3657	Stratovolcano	3.0	3.0
87	Indonesia	3657	Stratovolcano	70.0	3.0
88	Indonesia	3657	Stratovolcano	12.0	3.0
89	Indonesia	3657	Stratovolcano	119.0	2.0
90	Indonesia	3657	Stratovolcano	1.0	3.0
98	Indonesia	2769	Stratovolcano	NaN	2.0
99	Indonesia	1711	Caldera	2.0	2.0

100	Indonesia	2997	Stratovolcano	1028.0	5.0
101	Indonesia	2997	Stratovolcano	120.0	3.0
102	Indonesia	2850	Stratovolcano	60000.0	7.0
103	Indonesia	618	Stratovolcano	2.0	3.0
104	Indonesia	875	Stratovolcano	226.0	3.0
105	Indonesia	875	Stratovolcano	1.0	2.0
107	Indonesia	875	Stratovolcano	NaN	2.0
108	Indonesia	1703	Stratovolcano	2.0	2.0
109	Indonesia	1703	Stratovolcano	1.0	2.0
111	Indonesia	583	Complex volcano	NaN	2.0
113	Indonesia	583	Complex volcano	NaN	1.0
115	Pacific Ocean	728	Stratovolcano	NaN	4.0
117	Pacific Ocean	608	Stratovolcano	NaN	4.0
120	Pacific Ocean	596	Caldera	NaN	3.0
121	Pacific Ocean	596	Caldera	NaN	2.0
122	Pacific Ocean	596	Caldera	4.0	3.0
123	Indonesia	404	Stratovolcano	NaN	4.0
124	Indonesia	1785	Stratovolcano	118.0	2.0
125	Indonesia	1299	Stratovolcano	1.0	2.0
127	Indonesia	725	Stratovolcano	400.0	2.0
128	Indonesia	1797	Stratovolcano	1.0	2.0
129	Indonesia	1797	Stratovolcano	4.0	3.0
130	Indonesia	1797	Stratovolcano	1.0	2.0
131	Indonesia	-5	Submarine volcano	NaN	2.0
132	Indonesia	1318	Stratovolcano	3000.0	3.0
133	Indonesia	1318	Stratovolcano	953.0	3.0
134	Indonesia	1318	Stratovolcano	2806.0	3.0
135	Indonesia	1318	Stratovolcano	1532.0	3.0
136	Indonesia	1318	Stratovolcano	138.0	2.0
137	Indonesia	1318	Stratovolcano	39.0	4.0
139	Indonesia	1635	Stratovolcano	NaN	5.0
140	Indonesia	1714	Stratovolcano	NaN	3.0
141	Indonesia	1714	Stratovolcano	35.0	3.0
142	Indonesia	1714	Stratovolcano	NaN	2.0
144	Indonesia	1714	Stratovolcano	4.0	2.0
145	Indonesia	1714	Stratovolcano	NaN	3.0
146	Indonesia	1714	Stratovolcano	1.0	2.0
147	Indonesia	1714	Stratovolcano	5.0	2.0
149	Indonesia	1357	Stratovolcano	NaN	4.0
151	Indonesia	1357	Stratovolcano	326.0	4.0
152	Indonesia	1357	Stratovolcano	NaN	2.0
153	Philippines	2790	Stratovolcano	NaN	2.0
154	Philippines	1552	Stratovolcano	NaN	2.0
155	Philippines	1552	Stratovolcano	68.0	3.0
156	Philippines	2422	Stratovolcano	3.0	2.0
157	Philippines	1535	Stratovolcano	9.0	2.0
158	Philippines	1535	Stratovolcano	NaN	2.0
159	Philippines	2462	Stratovolcano	49.0	3.0
160	Philippines	2462	Stratovolcano	NaN	2.0
161	Philippines	2462	Stratovolcano	1200.0	4.0
162	Philippines	2462	Stratovolcano	34.0	3.0
164	Philippines	2462	Stratovolcano	3.0	3.0
165	Philippines	2462	Stratovolcano	15.0	3.0
166	Philippines	2462	Stratovolcano	350.0	3.0
168	Philippines	2462	Stratovolcano	NaN	2.0
169	Philippines	2462	Stratovolcano	NaN	2.0

170	Philippines	2462	Stratovolcano	6.0	3.0
171	Philippines	2462	Stratovolcano	1.0	3.0
173	Philippines	311	Stratovolcano	NaN	4.0
174	Philippines	311	Stratovolcano	NaN	3.0
175	Philippines	311	Stratovolcano	12.0	4.0
176	Philippines	311	Stratovolcano	NaN	2.0
177	Philippines	311	Stratovolcano	1335.0	3.0
178	Philippines	311	Stratovolcano	355.0	4.0
179	Philippines	228	Compound volcano	3.0	2.0
180	Taiwan	0	Submarine volcano	NaN	2.0
182	Japan	657	Stratovolcano	8.0	4.0
184	Japan	1117	Stratovolcano	NaN	3.0
185	Japan	1117	Stratovolcano	NaN	5.0
186	Japan	1117	Stratovolcano	153.0	4.0
187	Japan	1117	Stratovolcano	38.0	4.0
188	Japan	1117	Stratovolcano	63.0	4.0
190	Japan	1117	Stratovolcano	NaN	3.0
191	Japan	1700	Shield volcano	NaN	3.0
192	Japan	1700	Shield volcano	1.0	3.0
193	Japan	1700	Shield volcano	1.0	3.0
194	Japan	1483	Complex volcano	15000.0	2.0
195	Japan	1483	Complex volcano	43.0	1.0

deaths

0	NaN
1	NaN
3	NaN
5	NaN
6	NaN
8	NaN
10	NaN
11	1.0
12	1.0
13	1.0
14	1.0
15	1.0
16	2.0
17	2.0
19	2.0
20	3.0
22	4.0
23	4.0
24	4.0
25	5.0
26	4.0
27	5.0
28	9.0
29	9.0
30	2.0
31	11.0
32	NaN
33	20.0
34	1.0
35	24.0
36	25.0
37	NaN

38	27.0
39	37.0
41	48.0
42	50.0
43	59.0
44	117.0
45	149.0
47	3000.0
48	100.0
49	32.0
50	30.0
51	200.0
53	35.0
54	1369.0
55	64.0
56	6.0
57	3.0
58	1.0
59	66.0
65	NaN
66	NaN
67	22.0
68	54.0
69	NaN
70	NaN
71	5110.0
72	7.0
73	215.0
74	35.0
76	74.0
77	80.0
78	221.0
79	NaN
80	NaN
81	NaN
83	NaN
84	6.0
85	1.0
86	3.0
87	70.0
88	12.0
89	119.0
90	1.0
98	NaN
99	2.0
100	1028.0
101	120.0
102	11000.0
103	2.0
104	98.0
105	1.0
107	NaN
108	2.0
109	1.0
111	NaN
113	NaN

115	NaN
117	NaN
120	NaN
121	NaN
122	4.0
123	NaN
124	NaN
125	1.0
127	NaN
128	1.0
129	4.0
130	1.0
131	NaN
132	3000.0
133	953.0
134	2806.0
135	1532.0
136	NaN
137	39.0
139	NaN
140	NaN
141	35.0
142	NaN
144	4.0
145	NaN
146	1.0
147	5.0
149	NaN
151	326.0
152	NaN
153	NaN
154	NaN
155	68.0
156	3.0
157	NaN
158	NaN
159	49.0
160	NaN
161	1200.0
162	34.0
164	3.0
165	15.0
166	350.0
168	NaN
169	NaN
170	6.0
171	1.0
173	NaN
174	NaN
175	NaN
176	NaN
177	1335.0
178	355.0
179	NaN
180	NaN
182	8.0

184	NaN
185	NaN
186	153.0
187	23.0
188	28.0
190	NaN
191	NaN
192	1.0
193	1.0
194	NaN
195	43.0

1.2.2: *In one line of code, **change** the **index column** of the dataframe so that it has **1-based indexing**.*

```
In [36]: new_df.index = range(1, len(new_df)+1)

print(new_df)
```

	id	year	month	day	tsunamiEventId	earthquakeEventId	\
1	1	1169	2.0	4.0	2852.0	421.0	
2	2	1329	7.0	15.0	NaN	NaN	
3	4	1888	3.0	13.0	1175.0	NaN	
4	6	1832	11.0	1.0	NaN	NaN	
5	7	1977	1.0	10.0	NaN	NaN	
6	9	1779	8.0	8.0	NaN	NaN	
7	11	1907	8.0	4.0	NaN	NaN	
8	12	1905	3.0	10.0	NaN	NaN	
9	13	1986	7.0	24.0	NaN	NaN	
10	14	1536	3.0	26.0	NaN	NaN	
11	15	1904	2.0	25.0	NaN	NaN	
12	16	1878	2.0	4.0	1095.0	6195.0	
13	17	1737	5.0	20.0	NaN	NaN	
14	18	1917	4.0	1.0	NaN	NaN	
15	20	1979	3.0	8.0	NaN	NaN	
16	21	1970	3.0	2.0	NaN	NaN	
17	23	1682	8.0	12.0	NaN	NaN	
18	24	1805	8.0	11.0	NaN	NaN	
19	25	1919	5.0	22.0	1476.0	NaN	
20	26	1930	9.0	11.0	4218.0	NaN	
21	27	1903	8.0	30.0	NaN	NaN	
22	28	1928	11.0	8.0	NaN	NaN	
23	29	1872	4.0	24.0	NaN	NaN	
24	30	1979	9.0	12.0	NaN	NaN	
25	31	1987	4.0	17.0	NaN	NaN	
26	32	1914	9.0	10.0	NaN	NaN	
27	33	1886	8.0	31.0	NaN	NaN	
28	34	1912	12.0	3.0	NaN	NaN	
29	35	1895	6.0	17.0	NaN	NaN	
30	36	1538	9.0	29.0	NaN	NaN	
31	37	1853	6.0	24.0	NaN	NaN	
32	38	1954	8.0	3.0	NaN	NaN	
33	39	1944	3.0	27.0	NaN	NaN	
34	40	1984	8.0	15.0	5807.0	NaN	
35	42	1926	1.0	23.0	NaN	NaN	
36	43	1631	2.0	14.0	NaN	920.0	
37	44	1843	11.0	25.0	NaN	NaN	
38	45	1944	12.0	4.0	NaN	NaN	
39	49	1979	2.0	20.0	NaN	NaN	
40	64	1672	8.0	4.0	NaN	NaN	
41	69	1822	12.0	27.0	NaN	NaN	
42	71	1832	12.0	25.0	NaN	NaN	
43	81	1872	4.0	15.0	NaN	NaN	
44	82	1872	11.0	3.0	NaN	NaN	
45	92	1920	7.0	25.0	NaN	NaN	
46	94	1930	12.0	18.0	NaN	NaN	
47	100	1954	1.0	18.0	NaN	NaN	
48	104	1961	5.0	8.0	NaN	NaN	
49	105	1969	1.0	7.0	NaN	NaN	
50	111	1986	12.0	31.0	NaN	NaN	
51	112	1994	11.0	22.0	NaN	NaN	
52	128	1716	7.0	20.0	NaN	NaN	
53	132	1826	10.0	13.0	NaN	NaN	
54	133	1848	5.0	16.0	NaN	NaN	
55	135	1864	1.0	4.0	NaN	NaN	

56	136	1875	1.0	29.0	NaN	NaN
57	137	1901	5.0	22.0	NaN	NaN
58	138	1919	5.0	19.0	NaN	NaN
59	140	1951	8.0	31.0	NaN	NaN
60	141	1966	4.0	26.0	NaN	NaN
61	144	1990	2.0	10.0	NaN	NaN
62	167	1885	4.0	18.0	NaN	NaN
63	176	1895	5.0	22.0	NaN	NaN
64	191	1909	8.0	29.0	NaN	NaN
65	193	1911	11.0	8.0	NaN	NaN
66	195	1913	6.0	23.0	NaN	NaN
67	196	1941	9.0	21.0	NaN	NaN
68	199	1946	10.0	29.0	NaN	NaN
69	200	1950	8.0	28.0	NaN	NaN
70	201	1963	5.0	5.0	NaN	NaN
71	202	1967	8.0	31.0	NaN	NaN
72	203	1985	5.0	10.0	NaN	NaN
73	204	1978	9.0	19.0	NaN	NaN
74	205	1976	8.0	31.0	NaN	NaN
75	206	1981	3.0	29.0	NaN	NaN
76	333	1817	1.0	24.0	NaN	NaN
77	345	1963	9.0	5.0	NaN	NaN
78	351	1963	3.0	18.0	1942.0	NaN
79	353	1963	5.0	16.0	NaN	4292.0
80	367	1815	4.0	10.0	613.0	NaN
81	389	1969	1.0	28.0	NaN	NaN
82	394	1928	8.0	4.0	1607.0	NaN
83	395	1964	1.0	1.0	NaN	NaN
84	398	1981	9.0	5.0	NaN	NaN
85	410	1869	7.0	7.0	NaN	NaN
86	411	1907	9.0	28.0	NaN	NaN
87	441	1948	4.0	7.0	NaN	NaN
88	447	1983	8.0	17.0	4702.0	NaN
89	457	1659	11.0	11.0	285.0	NaN
90	466	1692	6.0	4.0	NaN	NaN
91	477	1694	11.0	20.0	NaN	NaN
92	482	1820	6.0	11.0	NaN	NaN
93	486	1988	5.0	9.0	NaN	NaN
94	488	1983	7.0	23.0	NaN	NaN
95	489	1845	2.0	8.0	741.0	1865.0
96	525	1958	7.0	12.0	NaN	NaN
97	533	1871	3.0	3.0	1022.0	NaN
98	552	1940	6.0	20.0	NaN	NaN
99	561	1974	2.0	11.0	NaN	NaN
100	562	1976	9.0	15.0	NaN	NaN
101	569	1889	9.0	6.0	1181.0	6309.0
102	578	1711	12.0	11.0	NaN	NaN
103	580	1812	8.0	6.0	NaN	NaN
104	581	1856	3.0	2.0	859.0	1995.0
105	584	1892	6.0	7.0	1197.0	NaN
106	585	1913	3.0	14.0	1407.0	2973.0
107	589	1966	8.0	12.0	NaN	NaN
108	599	1673	5.0	20.0	312.0	NaN
109	607	1608	7.0	18.0	222.0	NaN
110	617	1772	5.0	9.0	504.0	NaN
111	618	1773	2.0	2.0	NaN	NaN

112	627	1838	2.0	26.0	NaN	NaN
113	629	1840	2.0	2.0	718.0	NaN
114	643	1871	8.0	7.0	NaN	NaN
115	657	1962	12.0	31.0	NaN	NaN
116	664	1646	7.0	19.0	5754.0	10519.0
117	666	1861	12.0	29.0	NaN	NaN
118	669	1890	6.0	29.0	NaN	NaN
119	677	1873	1.0	16.0	NaN	NaN
120	679	1871	4.0	30.0	1020.0	2179.0
121	680	1950	9.0	15.0	NaN	NaN
122	693	1996	8.0	10.0	NaN	NaN
123	697	1933	12.0	25.0	1666.0	NaN
124	698	1978	7.0	29.0	NaN	NaN
125	704	1766	7.0	20.0	NaN	NaN
126	706	1800	10.0	30.0	NaN	NaN
127	707	1814	2.0	1.0	NaN	NaN
128	712	1853	7.0	13.0	NaN	NaN
129	716	1871	12.0	8.0	NaN	NaN
130	724	1887	3.0	9.0	NaN	NaN
131	730	1897	5.0	23.0	NaN	NaN
132	733	1938	6.0	5.0	NaN	NaN
133	735	1947	1.0	7.0	NaN	NaN
134	736	1968	4.0	21.0	NaN	NaN
135	739	1984	9.0	9.0	NaN	NaN
136	752	1716	9.0	24.0	387.0	1205.0
137	753	1749	8.0	11.0	431.0	1304.0
138	754	1754	11.0	28.0	2853.0	1328.0
139	756	1874	7.0	19.0	NaN	NaN
140	761	1911	1.0	30.0	1392.0	8391.0
141	762	1965	9.0	28.0	1977.0	NaN
142	777	1969	3.0	21.0	2872.0	NaN
143	787	1853	10.0	29.0	809.0	NaN
144	823	1933	12.0	24.0	NaN	NaN
145	844	766	7.0	20.0	61.0	7341.0
146	846	1471	11.0	3.0	NaN	NaN
147	858	1779	11.0	8.0	NaN	NaN
148	859	1781	4.0	11.0	529.0	NaN
149	867	1914	1.0	12.0	1412.0	2994.0
150	878	1955	10.0	13.0	NaN	NaN
151	904	1566	10.0	31.0	NaN	NaN
152	915	1716	11.0	9.0	NaN	NaN
153	916	1717	2.0	7.0	NaN	NaN
154	932	1792	5.0	21.0	568.0	1515.0
155	933	1991	6.0	3.0	NaN	NaN

	volcanoLocationId	volcanoLocationNewNum	name \
1	10106	211060	Etna
2	10106	211060	Etna
3	50107	251070	Ritter Island
4	10106	211060	Etna
5	20303	223030	Nyiragongo
6	10102	211020	Vesuvius
7	201112	221112	Alayta
8	10102	211020	Vesuvius
9	10104	211040	Stromboli
10	10106	211060	Etna

11	30301	233010	Karthala
12	50214	252140	Rabaul
13	10102	211020	Vesuvius
14	40105	241050	Okataina
15	50103	251030	Karkar
16	10101	211010	Campi Flegrei
17	10102	211020	Vesuvius
18	10102	211020	Vesuvius
19	10104	211040	Stromboli
20	10104	211040	Stromboli
21	40105	241050	Okataina
22	10106	211060	Etna
23	10102	211020	Vesuvius
24	10106	211060	Etna
25	10106	211060	Etna
26	40104	241040	Whakaari/White Island
27	40311	243110	Niuafu'ou
28	20302	223020	Nyamulagira
29	50103	251030	Karkar
30	10101	211010	Campi Flegrei
31	40311	243110	Niuafu'ou
32	50101	251010	Bam
33	10102	211020	Vesuvius
34	20403	224030	Oku Volcanic Field
35	10204	212040	Santorini
36	201141	221141	Dama Ali
37	10106	211060	Etna
38	60320	263200	Dieng Volcanic Complex
39	60320	263200	Dieng Volcanic Complex
40	60325	263250	Merapi
41	60325	263250	Merapi
42	60325	263250	Merapi
43	60325	263250	Merapi
44	60325	263250	Merapi
45	60325	263250	Merapi
46	60325	263250	Merapi
47	60325	263250	Merapi
48	60325	263250	Merapi
49	60325	263250	Merapi
50	60325	263250	Merapi
51	60325	263250	Merapi
52	60328	263280	Kelud
53	60328	263280	Kelud
54	60328	263280	Kelud
55	60328	263280	Kelud
56	60328	263280	Kelud
57	60328	263280	Kelud
58	60328	263280	Kelud
59	60328	263280	Kelud
60	60328	263280	Kelud
61	60328	263280	Kelud
62	60330	263300	Semeru
63	60330	263300	Semeru
64	60330	263300	Semeru
65	60330	263300	Semeru
66	60330	263300	Semeru

67	60330	263300	Semeru
68	60330	263300	Semeru
69	60330	263300	Semeru
70	60330	263300	Semeru
71	60330	263300	Semeru
72	60330	263300	Semeru
73	60330	263300	Semeru
74	60330	263300	Semeru
75	60330	263300	Semeru
76	60335	263350	Ijen
77	60401	264010	Batur
78	60402	264020	Agung
79	60402	264020	Agung
80	60404	264040	Tambora
81	60411	264110	Iya
82	60415	264150	Paluweh
83	60415	264150	Paluweh
84	60415	264150	Paluweh
85	60418	264180	Lewotobi
86	60418	264180	Lewotobi
87	60425	264250	Iliwerung
88	60425	264250	Iliwerung
89	60505	265050	Teon
90	60507	265070	Serua
91	60509	265090	Banda Api
92	60509	265090	Banda Api
93	60509	265090	Banda Api
94	60601	266010	Colo
95	60603	266030	Soputan
96	60611	266110	Mahawu
97	60701	267010	Ruang
98	60702	267020	Karangetang
99	60702	267020	Karangetang
100	60702	267020	Karangetang
101	60703	267030	Banua Wuhu
102	60704	267040	Awu
103	60704	267040	Awu
104	60704	267040	Awu
105	60704	267040	Awu
106	60704	267040	Awu
107	60704	267040	Awu
108	60804	268040	Gamkonora
109	60806	268060	Gamalama
110	60806	268060	Gamalama
111	60806	268060	Gamalama
112	60806	268060	Gamalama
113	60806	268060	Gamalama
114	60806	268060	Gamalama
115	60806	268060	Gamalama
116	60807	268070	Kie Besi
117	60807	268070	Kie Besi
118	60807	268070	Kie Besi
119	70106	271060	Ragang
120	70108	271080	Camiguin
121	70108	271080	Camiguin
122	70202	272020	Kanlaon

123	70301	273010	Bulusan
124	70301	273010	Bulusan
125	70303	273030	Mayon
126	70303	273030	Mayon
127	70303	273030	Mayon
128	70303	273030	Mayon
129	70303	273030	Mayon
130	70303	273030	Mayon
131	70303	273030	Mayon
132	70303	273030	Mayon
133	70303	273030	Mayon
134	70303	273030	Mayon
135	70303	273030	Mayon
136	70307	273070	Taal
137	70307	273070	Taal
138	70307	273070	Taal
139	70307	273070	Taal
140	70307	273070	Taal
141	70307	273070	Taal
142	70402	274020	Didicas
143	80103	281030	Unnamed
144	80205	282050	Kuchinoerabujima
145	80208	282080	Aira
146	80208	282080	Aira
147	80208	282080	Aira
148	80208	282080	Aira
149	80208	282080	Aira
150	80208	282080	Aira
151	80209	282090	Kirishimayama
152	80209	282090	Kirishimayama
153	80209	282090	Kirishimayama
154	80210	282100	Unzendake
155	80210	282100	Unzendake

	country	elevation	morphology	deathsTotal	vei \
1	Italy	3357	Stratovolcano	16000.0	NaN
2	Italy	3357	Stratovolcano	NaN	3.0
3	Papua New Guinea	75	Stratovolcano	NaN	2.0
4	Italy	3357	Stratovolcano	NaN	2.0
5	Congo, DRC	3470	Stratovolcano	NaN	1.0
6	Italy	1281	Complex volcano	NaN	3.0
7	Ethiopia	1496	Shield volcano	NaN	2.0
8	Italy	1281	Complex volcano	1.0	2.0
9	Italy	924	Stratovolcano	1.0	2.0
10	Italy	3357	Stratovolcano	1.0	3.0
11	Comoros	2361	Shield volcano	1.0	2.0
12	Papua New Guinea	688	Pyroclastic shield	1.0	3.0
13	Italy	1281	Complex volcano	2.0	3.0
14	New Zealand	1111	Lava dome	2.0	1.0
15	Papua New Guinea	1839	Stratovolcano	2.0	2.0
16	Italy	458	Caldera	3.0	NaN
17	Italy	1281	Complex volcano	4.0	3.0
18	Italy	1281	Complex volcano	4.0	2.0
19	Italy	924	Stratovolcano	4.0	3.0
20	Italy	924	Stratovolcano	6.0	3.0
21	New Zealand	1111	Lava dome	4.0	1.0

22	Italy	3357	Stratovolcano	5.0	1.0
23	Italy	1281	Complex volcano	9.0	3.0
24	Italy	3357	Stratovolcano	9.0	3.0
25	Italy	3357	Stratovolcano	2.0	2.0
26	New Zealand	294	Stratovolcano	11.0	NaN
27	Tonga	260	Shield volcano	NaN	4.0
28	Congo, DRC	3058	Shield volcano	20.0	3.0
29	Papua New Guinea	1839	Stratovolcano	21.0	2.0
30	Italy	458	Caldera	24.0	3.0
31	Tonga	260	Shield volcano	25.0	0.0
32	Papua New Guinea	685	Stratovolcano	25.0	2.0
33	Italy	1281	Complex volcano	27.0	3.0
34	Cameroon	3011	Maar	37.0	NaN
35	Greece	367	Shield volcano	48.0	2.0
36	Ethiopia	1068	Shield volcano	50.0	NaN
37	Italy	3357	Stratovolcano	59.0	2.0
38	Indonesia	2565	Complex volcano	117.0	2.0
39	Indonesia	2565	Complex volcano	149.0	1.0
40	Indonesia	2910	Stratovolcano	3000.0	3.0
41	Indonesia	2910	Stratovolcano	100.0	3.0
42	Indonesia	2910	Stratovolcano	32.0	2.0
43	Indonesia	2910	Stratovolcano	30.0	4.0
44	Indonesia	2910	Stratovolcano	200.0	4.0
45	Indonesia	2910	Stratovolcano	35.0	2.0
46	Indonesia	2910	Stratovolcano	1369.0	3.0
47	Indonesia	2910	Stratovolcano	64.0	2.0
48	Indonesia	2910	Stratovolcano	6.0	3.0
49	Indonesia	2910	Stratovolcano	3.0	2.0
50	Indonesia	2910	Stratovolcano	1.0	2.0
51	Indonesia	2910	Stratovolcano	66.0	3.0
52	Indonesia	1730	Stratovolcano	NaN	2.0
53	Indonesia	1730	Stratovolcano	NaN	4.0
54	Indonesia	1730	Stratovolcano	22.0	3.0
55	Indonesia	1730	Stratovolcano	54.0	2.0
56	Indonesia	1730	Stratovolcano	NaN	NaN
57	Indonesia	1730	Stratovolcano	NaN	3.0
58	Indonesia	1730	Stratovolcano	5110.0	4.0
59	Indonesia	1730	Stratovolcano	7.0	4.0
60	Indonesia	1730	Stratovolcano	215.0	4.0
61	Indonesia	1730	Stratovolcano	35.0	4.0
62	Indonesia	3657	Stratovolcano	74.0	2.0
63	Indonesia	3657	Stratovolcano	80.0	2.0
64	Indonesia	3657	Stratovolcano	221.0	2.0
65	Indonesia	3657	Stratovolcano	NaN	3.0
66	Indonesia	3657	Stratovolcano	NaN	2.0
67	Indonesia	3657	Stratovolcano	NaN	2.0
68	Indonesia	3657	Stratovolcano	NaN	2.0
69	Indonesia	3657	Stratovolcano	6.0	1.0
70	Indonesia	3657	Stratovolcano	1.0	2.0
71	Indonesia	3657	Stratovolcano	3.0	3.0
72	Indonesia	3657	Stratovolcano	70.0	3.0
73	Indonesia	3657	Stratovolcano	12.0	3.0
74	Indonesia	3657	Stratovolcano	119.0	2.0
75	Indonesia	3657	Stratovolcano	1.0	3.0
76	Indonesia	2769	Stratovolcano	NaN	2.0
77	Indonesia	1711	Caldera	2.0	2.0

78	Indonesia	2997	Stratovolcano	1028.0	5.0
79	Indonesia	2997	Stratovolcano	120.0	3.0
80	Indonesia	2850	Stratovolcano	60000.0	7.0
81	Indonesia	618	Stratovolcano	2.0	3.0
82	Indonesia	875	Stratovolcano	226.0	3.0
83	Indonesia	875	Stratovolcano	1.0	2.0
84	Indonesia	875	Stratovolcano	NaN	2.0
85	Indonesia	1703	Stratovolcano	2.0	2.0
86	Indonesia	1703	Stratovolcano	1.0	2.0
87	Indonesia	583	Complex volcano	NaN	2.0
88	Indonesia	583	Complex volcano	NaN	1.0
89	Pacific Ocean	728	Stratovolcano	NaN	4.0
90	Pacific Ocean	608	Stratovolcano	NaN	4.0
91	Pacific Ocean	596	Caldera	NaN	3.0
92	Pacific Ocean	596	Caldera	NaN	2.0
93	Pacific Ocean	596	Caldera	4.0	3.0
94	Indonesia	404	Stratovolcano	NaN	4.0
95	Indonesia	1785	Stratovolcano	118.0	2.0
96	Indonesia	1299	Stratovolcano	1.0	2.0
97	Indonesia	725	Stratovolcano	400.0	2.0
98	Indonesia	1797	Stratovolcano	1.0	2.0
99	Indonesia	1797	Stratovolcano	4.0	3.0
100	Indonesia	1797	Stratovolcano	1.0	2.0
101	Indonesia	-5	Submarine volcano	NaN	2.0
102	Indonesia	1318	Stratovolcano	3000.0	3.0
103	Indonesia	1318	Stratovolcano	953.0	3.0
104	Indonesia	1318	Stratovolcano	2806.0	3.0
105	Indonesia	1318	Stratovolcano	1532.0	3.0
106	Indonesia	1318	Stratovolcano	138.0	2.0
107	Indonesia	1318	Stratovolcano	39.0	4.0
108	Indonesia	1635	Stratovolcano	NaN	5.0
109	Indonesia	1714	Stratovolcano	NaN	3.0
110	Indonesia	1714	Stratovolcano	35.0	3.0
111	Indonesia	1714	Stratovolcano	NaN	2.0
112	Indonesia	1714	Stratovolcano	4.0	2.0
113	Indonesia	1714	Stratovolcano	NaN	3.0
114	Indonesia	1714	Stratovolcano	1.0	2.0
115	Indonesia	1714	Stratovolcano	5.0	2.0
116	Indonesia	1357	Stratovolcano	NaN	4.0
117	Indonesia	1357	Stratovolcano	326.0	4.0
118	Indonesia	1357	Stratovolcano	NaN	2.0
119	Philippines	2790	Stratovolcano	NaN	2.0
120	Philippines	1552	Stratovolcano	NaN	2.0
121	Philippines	1552	Stratovolcano	68.0	3.0
122	Philippines	2422	Stratovolcano	3.0	2.0
123	Philippines	1535	Stratovolcano	9.0	2.0
124	Philippines	1535	Stratovolcano	NaN	2.0
125	Philippines	2462	Stratovolcano	49.0	3.0
126	Philippines	2462	Stratovolcano	NaN	2.0
127	Philippines	2462	Stratovolcano	1200.0	4.0
128	Philippines	2462	Stratovolcano	34.0	3.0
129	Philippines	2462	Stratovolcano	3.0	3.0
130	Philippines	2462	Stratovolcano	15.0	3.0
131	Philippines	2462	Stratovolcano	350.0	3.0
132	Philippines	2462	Stratovolcano	NaN	2.0
133	Philippines	2462	Stratovolcano	NaN	2.0

134	Philippines	2462	Stratovolcano	6.0	3.0
135	Philippines	2462	Stratovolcano	1.0	3.0
136	Philippines	311	Stratovolcano	NaN	4.0
137	Philippines	311	Stratovolcano	NaN	3.0
138	Philippines	311	Stratovolcano	12.0	4.0
139	Philippines	311	Stratovolcano	NaN	2.0
140	Philippines	311	Stratovolcano	1335.0	3.0
141	Philippines	311	Stratovolcano	355.0	4.0
142	Philippines	228	Compound volcano	3.0	2.0
143	Taiwan	0	Submarine volcano	NaN	2.0
144	Japan	657	Stratovolcano	8.0	4.0
145	Japan	1117	Stratovolcano	NaN	3.0
146	Japan	1117	Stratovolcano	NaN	5.0
147	Japan	1117	Stratovolcano	153.0	4.0
148	Japan	1117	Stratovolcano	38.0	4.0
149	Japan	1117	Stratovolcano	63.0	4.0
150	Japan	1117	Stratovolcano	NaN	3.0
151	Japan	1700	Shield volcano	NaN	3.0
152	Japan	1700	Shield volcano	1.0	3.0
153	Japan	1700	Shield volcano	1.0	3.0
154	Japan	1483	Complex volcano	15000.0	2.0
155	Japan	1483	Complex volcano	43.0	1.0

deaths

1	NaN
2	NaN
3	NaN
4	NaN
5	NaN
6	NaN
7	NaN
8	1.0
9	1.0
10	1.0
11	1.0
12	1.0
13	2.0
14	2.0
15	2.0
16	3.0
17	4.0
18	4.0
19	4.0
20	5.0
21	4.0
22	5.0
23	9.0
24	9.0
25	2.0
26	11.0
27	NaN
28	20.0
29	1.0
30	24.0
31	25.0
32	NaN

33	27.0
34	37.0
35	48.0
36	50.0
37	59.0
38	117.0
39	149.0
40	3000.0
41	100.0
42	32.0
43	30.0
44	200.0
45	35.0
46	1369.0
47	64.0
48	6.0
49	3.0
50	1.0
51	66.0
52	NaN
53	NaN
54	22.0
55	54.0
56	NaN
57	NaN
58	5110.0
59	7.0
60	215.0
61	35.0
62	74.0
63	80.0
64	221.0
65	NaN
66	NaN
67	NaN
68	NaN
69	6.0
70	1.0
71	3.0
72	70.0
73	12.0
74	119.0
75	1.0
76	NaN
77	2.0
78	1028.0
79	120.0
80	11000.0
81	2.0
82	98.0
83	1.0
84	NaN
85	2.0
86	1.0
87	NaN
88	NaN

89	NaN
90	NaN
91	NaN
92	NaN
93	4.0
94	NaN
95	NaN
96	1.0
97	NaN
98	1.0
99	4.0
100	1.0
101	NaN
102	3000.0
103	953.0
104	2806.0
105	1532.0
106	NaN
107	39.0
108	NaN
109	NaN
110	35.0
111	NaN
112	4.0
113	NaN
114	1.0
115	5.0
116	NaN
117	326.0
118	NaN
119	NaN
120	NaN
121	68.0
122	3.0
123	NaN
124	NaN
125	49.0
126	NaN
127	1200.0
128	34.0
129	3.0
130	15.0
131	350.0
132	NaN
133	NaN
134	6.0
135	1.0
136	NaN
137	NaN
138	NaN
139	NaN
140	1335.0
141	355.0
142	NaN
143	NaN
144	8.0

145	NaN
146	NaN
147	153.0
148	23.0
149	28.0
150	NaN
151	NaN
152	1.0
153	1.0
154	NaN
155	43.0

The **deathsTotal** and **deaths** columns have approximations of the same data with alternating NaNs in each.

1.2.3: In one line of code, make a **new column** called '**totalDeaths**' that takes the **max** of the values given between those **two** *columns.

- If there is **NaN** in **one column** and a **numerical** value in the **other**, it will **take the numerical value**.
- **Only** if there are **NaNs** in **both columns**, the **new column** will have **NaN**.

```
In [37]: new_df['totalDeaths'] = new_df[['deathsTotal', 'deaths']].max(axis=1, skipna=True)
print(new_df)
```

	id	year	month	day	tsunamiEventId	earthquakeEventId	\
1	1	1169	2.0	4.0	2852.0	421.0	
2	2	1329	7.0	15.0	NaN	NaN	
3	4	1888	3.0	13.0	1175.0	NaN	
4	6	1832	11.0	1.0	NaN	NaN	
5	7	1977	1.0	10.0	NaN	NaN	
6	9	1779	8.0	8.0	NaN	NaN	
7	11	1907	8.0	4.0	NaN	NaN	
8	12	1905	3.0	10.0	NaN	NaN	
9	13	1986	7.0	24.0	NaN	NaN	
10	14	1536	3.0	26.0	NaN	NaN	
11	15	1904	2.0	25.0	NaN	NaN	
12	16	1878	2.0	4.0	1095.0	6195.0	
13	17	1737	5.0	20.0	NaN	NaN	
14	18	1917	4.0	1.0	NaN	NaN	
15	20	1979	3.0	8.0	NaN	NaN	
16	21	1970	3.0	2.0	NaN	NaN	
17	23	1682	8.0	12.0	NaN	NaN	
18	24	1805	8.0	11.0	NaN	NaN	
19	25	1919	5.0	22.0	1476.0	NaN	
20	26	1930	9.0	11.0	4218.0	NaN	
21	27	1903	8.0	30.0	NaN	NaN	
22	28	1928	11.0	8.0	NaN	NaN	
23	29	1872	4.0	24.0	NaN	NaN	
24	30	1979	9.0	12.0	NaN	NaN	
25	31	1987	4.0	17.0	NaN	NaN	
26	32	1914	9.0	10.0	NaN	NaN	
27	33	1886	8.0	31.0	NaN	NaN	
28	34	1912	12.0	3.0	NaN	NaN	
29	35	1895	6.0	17.0	NaN	NaN	
30	36	1538	9.0	29.0	NaN	NaN	
31	37	1853	6.0	24.0	NaN	NaN	
32	38	1954	8.0	3.0	NaN	NaN	
33	39	1944	3.0	27.0	NaN	NaN	
34	40	1984	8.0	15.0	5807.0	NaN	
35	42	1926	1.0	23.0	NaN	NaN	
36	43	1631	2.0	14.0	NaN	920.0	
37	44	1843	11.0	25.0	NaN	NaN	
38	45	1944	12.0	4.0	NaN	NaN	
39	49	1979	2.0	20.0	NaN	NaN	
40	64	1672	8.0	4.0	NaN	NaN	
41	69	1822	12.0	27.0	NaN	NaN	
42	71	1832	12.0	25.0	NaN	NaN	
43	81	1872	4.0	15.0	NaN	NaN	
44	82	1872	11.0	3.0	NaN	NaN	
45	92	1920	7.0	25.0	NaN	NaN	
46	94	1930	12.0	18.0	NaN	NaN	
47	100	1954	1.0	18.0	NaN	NaN	
48	104	1961	5.0	8.0	NaN	NaN	
49	105	1969	1.0	7.0	NaN	NaN	
50	111	1986	12.0	31.0	NaN	NaN	
51	112	1994	11.0	22.0	NaN	NaN	
52	128	1716	7.0	20.0	NaN	NaN	
53	132	1826	10.0	13.0	NaN	NaN	
54	133	1848	5.0	16.0	NaN	NaN	
55	135	1864	1.0	4.0	NaN	NaN	

56	136	1875	1.0	29.0	NaN	NaN
57	137	1901	5.0	22.0	NaN	NaN
58	138	1919	5.0	19.0	NaN	NaN
59	140	1951	8.0	31.0	NaN	NaN
60	141	1966	4.0	26.0	NaN	NaN
61	144	1990	2.0	10.0	NaN	NaN
62	167	1885	4.0	18.0	NaN	NaN
63	176	1895	5.0	22.0	NaN	NaN
64	191	1909	8.0	29.0	NaN	NaN
65	193	1911	11.0	8.0	NaN	NaN
66	195	1913	6.0	23.0	NaN	NaN
67	196	1941	9.0	21.0	NaN	NaN
68	199	1946	10.0	29.0	NaN	NaN
69	200	1950	8.0	28.0	NaN	NaN
70	201	1963	5.0	5.0	NaN	NaN
71	202	1967	8.0	31.0	NaN	NaN
72	203	1985	5.0	10.0	NaN	NaN
73	204	1978	9.0	19.0	NaN	NaN
74	205	1976	8.0	31.0	NaN	NaN
75	206	1981	3.0	29.0	NaN	NaN
76	333	1817	1.0	24.0	NaN	NaN
77	345	1963	9.0	5.0	NaN	NaN
78	351	1963	3.0	18.0	1942.0	NaN
79	353	1963	5.0	16.0	NaN	4292.0
80	367	1815	4.0	10.0	613.0	NaN
81	389	1969	1.0	28.0	NaN	NaN
82	394	1928	8.0	4.0	1607.0	NaN
83	395	1964	1.0	1.0	NaN	NaN
84	398	1981	9.0	5.0	NaN	NaN
85	410	1869	7.0	7.0	NaN	NaN
86	411	1907	9.0	28.0	NaN	NaN
87	441	1948	4.0	7.0	NaN	NaN
88	447	1983	8.0	17.0	4702.0	NaN
89	457	1659	11.0	11.0	285.0	NaN
90	466	1692	6.0	4.0	NaN	NaN
91	477	1694	11.0	20.0	NaN	NaN
92	482	1820	6.0	11.0	NaN	NaN
93	486	1988	5.0	9.0	NaN	NaN
94	488	1983	7.0	23.0	NaN	NaN
95	489	1845	2.0	8.0	741.0	1865.0
96	525	1958	7.0	12.0	NaN	NaN
97	533	1871	3.0	3.0	1022.0	NaN
98	552	1940	6.0	20.0	NaN	NaN
99	561	1974	2.0	11.0	NaN	NaN
100	562	1976	9.0	15.0	NaN	NaN
101	569	1889	9.0	6.0	1181.0	6309.0
102	578	1711	12.0	11.0	NaN	NaN
103	580	1812	8.0	6.0	NaN	NaN
104	581	1856	3.0	2.0	859.0	1995.0
105	584	1892	6.0	7.0	1197.0	NaN
106	585	1913	3.0	14.0	1407.0	2973.0
107	589	1966	8.0	12.0	NaN	NaN
108	599	1673	5.0	20.0	312.0	NaN
109	607	1608	7.0	18.0	222.0	NaN
110	617	1772	5.0	9.0	504.0	NaN
111	618	1773	2.0	2.0	NaN	NaN

112	627	1838	2.0	26.0	NaN	NaN
113	629	1840	2.0	2.0	718.0	NaN
114	643	1871	8.0	7.0	NaN	NaN
115	657	1962	12.0	31.0	NaN	NaN
116	664	1646	7.0	19.0	5754.0	10519.0
117	666	1861	12.0	29.0	NaN	NaN
118	669	1890	6.0	29.0	NaN	NaN
119	677	1873	1.0	16.0	NaN	NaN
120	679	1871	4.0	30.0	1020.0	2179.0
121	680	1950	9.0	15.0	NaN	NaN
122	693	1996	8.0	10.0	NaN	NaN
123	697	1933	12.0	25.0	1666.0	NaN
124	698	1978	7.0	29.0	NaN	NaN
125	704	1766	7.0	20.0	NaN	NaN
126	706	1800	10.0	30.0	NaN	NaN
127	707	1814	2.0	1.0	NaN	NaN
128	712	1853	7.0	13.0	NaN	NaN
129	716	1871	12.0	8.0	NaN	NaN
130	724	1887	3.0	9.0	NaN	NaN
131	730	1897	5.0	23.0	NaN	NaN
132	733	1938	6.0	5.0	NaN	NaN
133	735	1947	1.0	7.0	NaN	NaN
134	736	1968	4.0	21.0	NaN	NaN
135	739	1984	9.0	9.0	NaN	NaN
136	752	1716	9.0	24.0	387.0	1205.0
137	753	1749	8.0	11.0	431.0	1304.0
138	754	1754	11.0	28.0	2853.0	1328.0
139	756	1874	7.0	19.0	NaN	NaN
140	761	1911	1.0	30.0	1392.0	8391.0
141	762	1965	9.0	28.0	1977.0	NaN
142	777	1969	3.0	21.0	2872.0	NaN
143	787	1853	10.0	29.0	809.0	NaN
144	823	1933	12.0	24.0	NaN	NaN
145	844	766	7.0	20.0	61.0	7341.0
146	846	1471	11.0	3.0	NaN	NaN
147	858	1779	11.0	8.0	NaN	NaN
148	859	1781	4.0	11.0	529.0	NaN
149	867	1914	1.0	12.0	1412.0	2994.0
150	878	1955	10.0	13.0	NaN	NaN
151	904	1566	10.0	31.0	NaN	NaN
152	915	1716	11.0	9.0	NaN	NaN
153	916	1717	2.0	7.0	NaN	NaN
154	932	1792	5.0	21.0	568.0	1515.0
155	933	1991	6.0	3.0	NaN	NaN

	volcanoLocationId	volcanoLocationNewNum	name \
1	10106	211060	Etna
2	10106	211060	Etna
3	50107	251070	Ritter Island
4	10106	211060	Etna
5	20303	223030	Nyiragongo
6	10102	211020	Vesuvius
7	201112	221112	Alayta
8	10102	211020	Vesuvius
9	10104	211040	Stromboli
10	10106	211060	Etna

11	30301	233010	Karthala
12	50214	252140	Rabaul
13	10102	211020	Vesuvius
14	40105	241050	Okataina
15	50103	251030	Karkar
16	10101	211010	Campi Flegrei
17	10102	211020	Vesuvius
18	10102	211020	Vesuvius
19	10104	211040	Stromboli
20	10104	211040	Stromboli
21	40105	241050	Okataina
22	10106	211060	Etna
23	10102	211020	Vesuvius
24	10106	211060	Etna
25	10106	211060	Etna
26	40104	241040	Whakaari/White Island
27	40311	243110	Niuafu'ou
28	20302	223020	Nyamulagira
29	50103	251030	Karkar
30	10101	211010	Campi Flegrei
31	40311	243110	Niuafu'ou
32	50101	251010	Bam
33	10102	211020	Vesuvius
34	20403	224030	Oku Volcanic Field
35	10204	212040	Santorini
36	201141	221141	Dama Ali
37	10106	211060	Etna
38	60320	263200	Dieng Volcanic Complex
39	60320	263200	Dieng Volcanic Complex
40	60325	263250	Merapi
41	60325	263250	Merapi
42	60325	263250	Merapi
43	60325	263250	Merapi
44	60325	263250	Merapi
45	60325	263250	Merapi
46	60325	263250	Merapi
47	60325	263250	Merapi
48	60325	263250	Merapi
49	60325	263250	Merapi
50	60325	263250	Merapi
51	60325	263250	Merapi
52	60328	263280	Kelud
53	60328	263280	Kelud
54	60328	263280	Kelud
55	60328	263280	Kelud
56	60328	263280	Kelud
57	60328	263280	Kelud
58	60328	263280	Kelud
59	60328	263280	Kelud
60	60328	263280	Kelud
61	60328	263280	Kelud
62	60330	263300	Semeru
63	60330	263300	Semeru
64	60330	263300	Semeru
65	60330	263300	Semeru
66	60330	263300	Semeru

67	60330	263300	Semeru
68	60330	263300	Semeru
69	60330	263300	Semeru
70	60330	263300	Semeru
71	60330	263300	Semeru
72	60330	263300	Semeru
73	60330	263300	Semeru
74	60330	263300	Semeru
75	60330	263300	Semeru
76	60335	263350	Ijen
77	60401	264010	Batur
78	60402	264020	Agung
79	60402	264020	Agung
80	60404	264040	Tambora
81	60411	264110	Iya
82	60415	264150	Paluweh
83	60415	264150	Paluweh
84	60415	264150	Paluweh
85	60418	264180	Lewotobi
86	60418	264180	Lewotobi
87	60425	264250	Iliwerung
88	60425	264250	Iliwerung
89	60505	265050	Teon
90	60507	265070	Serua
91	60509	265090	Banda Api
92	60509	265090	Banda Api
93	60509	265090	Banda Api
94	60601	266010	Colo
95	60603	266030	Soputan
96	60611	266110	Mahawu
97	60701	267010	Ruang
98	60702	267020	Karangetang
99	60702	267020	Karangetang
100	60702	267020	Karangetang
101	60703	267030	Banua Wuhu
102	60704	267040	Awu
103	60704	267040	Awu
104	60704	267040	Awu
105	60704	267040	Awu
106	60704	267040	Awu
107	60704	267040	Awu
108	60804	268040	Gamkonora
109	60806	268060	Gamalama
110	60806	268060	Gamalama
111	60806	268060	Gamalama
112	60806	268060	Gamalama
113	60806	268060	Gamalama
114	60806	268060	Gamalama
115	60806	268060	Gamalama
116	60807	268070	Kie Besi
117	60807	268070	Kie Besi
118	60807	268070	Kie Besi
119	70106	271060	Ragang
120	70108	271080	Camiguin
121	70108	271080	Camiguin
122	70202	272020	Kanlaon

123	70301	273010	Bulusan
124	70301	273010	Bulusan
125	70303	273030	Mayon
126	70303	273030	Mayon
127	70303	273030	Mayon
128	70303	273030	Mayon
129	70303	273030	Mayon
130	70303	273030	Mayon
131	70303	273030	Mayon
132	70303	273030	Mayon
133	70303	273030	Mayon
134	70303	273030	Mayon
135	70303	273030	Mayon
136	70307	273070	Taal
137	70307	273070	Taal
138	70307	273070	Taal
139	70307	273070	Taal
140	70307	273070	Taal
141	70307	273070	Taal
142	70402	274020	Didicas
143	80103	281030	Unnamed
144	80205	282050	Kuchinoerabujima
145	80208	282080	Aira
146	80208	282080	Aira
147	80208	282080	Aira
148	80208	282080	Aira
149	80208	282080	Aira
150	80208	282080	Aira
151	80209	282090	Kirishimayama
152	80209	282090	Kirishimayama
153	80209	282090	Kirishimayama
154	80210	282100	Unzendake
155	80210	282100	Unzendake

	country	elevation	morphology	deathsTotal	vei \
1	Italy	3357	Stratovolcano	16000.0	NaN
2	Italy	3357	Stratovolcano	NaN	3.0
3	Papua New Guinea	75	Stratovolcano	NaN	2.0
4	Italy	3357	Stratovolcano	NaN	2.0
5	Congo, DRC	3470	Stratovolcano	NaN	1.0
6	Italy	1281	Complex volcano	NaN	3.0
7	Ethiopia	1496	Shield volcano	NaN	2.0
8	Italy	1281	Complex volcano	1.0	2.0
9	Italy	924	Stratovolcano	1.0	2.0
10	Italy	3357	Stratovolcano	1.0	3.0
11	Comoros	2361	Shield volcano	1.0	2.0
12	Papua New Guinea	688	Pyroclastic shield	1.0	3.0
13	Italy	1281	Complex volcano	2.0	3.0
14	New Zealand	1111	Lava dome	2.0	1.0
15	Papua New Guinea	1839	Stratovolcano	2.0	2.0
16	Italy	458	Caldera	3.0	NaN
17	Italy	1281	Complex volcano	4.0	3.0
18	Italy	1281	Complex volcano	4.0	2.0
19	Italy	924	Stratovolcano	4.0	3.0
20	Italy	924	Stratovolcano	6.0	3.0
21	New Zealand	1111	Lava dome	4.0	1.0

22	Italy	3357	Stratovolcano	5.0	1.0
23	Italy	1281	Complex volcano	9.0	3.0
24	Italy	3357	Stratovolcano	9.0	3.0
25	Italy	3357	Stratovolcano	2.0	2.0
26	New Zealand	294	Stratovolcano	11.0	NaN
27	Tonga	260	Shield volcano	NaN	4.0
28	Congo, DRC	3058	Shield volcano	20.0	3.0
29	Papua New Guinea	1839	Stratovolcano	21.0	2.0
30	Italy	458	Caldera	24.0	3.0
31	Tonga	260	Shield volcano	25.0	0.0
32	Papua New Guinea	685	Stratovolcano	25.0	2.0
33	Italy	1281	Complex volcano	27.0	3.0
34	Cameroon	3011	Maar	37.0	NaN
35	Greece	367	Shield volcano	48.0	2.0
36	Ethiopia	1068	Shield volcano	50.0	NaN
37	Italy	3357	Stratovolcano	59.0	2.0
38	Indonesia	2565	Complex volcano	117.0	2.0
39	Indonesia	2565	Complex volcano	149.0	1.0
40	Indonesia	2910	Stratovolcano	3000.0	3.0
41	Indonesia	2910	Stratovolcano	100.0	3.0
42	Indonesia	2910	Stratovolcano	32.0	2.0
43	Indonesia	2910	Stratovolcano	30.0	4.0
44	Indonesia	2910	Stratovolcano	200.0	4.0
45	Indonesia	2910	Stratovolcano	35.0	2.0
46	Indonesia	2910	Stratovolcano	1369.0	3.0
47	Indonesia	2910	Stratovolcano	64.0	2.0
48	Indonesia	2910	Stratovolcano	6.0	3.0
49	Indonesia	2910	Stratovolcano	3.0	2.0
50	Indonesia	2910	Stratovolcano	1.0	2.0
51	Indonesia	2910	Stratovolcano	66.0	3.0
52	Indonesia	1730	Stratovolcano	NaN	2.0
53	Indonesia	1730	Stratovolcano	NaN	4.0
54	Indonesia	1730	Stratovolcano	22.0	3.0
55	Indonesia	1730	Stratovolcano	54.0	2.0
56	Indonesia	1730	Stratovolcano	NaN	NaN
57	Indonesia	1730	Stratovolcano	NaN	3.0
58	Indonesia	1730	Stratovolcano	5110.0	4.0
59	Indonesia	1730	Stratovolcano	7.0	4.0
60	Indonesia	1730	Stratovolcano	215.0	4.0
61	Indonesia	1730	Stratovolcano	35.0	4.0
62	Indonesia	3657	Stratovolcano	74.0	2.0
63	Indonesia	3657	Stratovolcano	80.0	2.0
64	Indonesia	3657	Stratovolcano	221.0	2.0
65	Indonesia	3657	Stratovolcano	NaN	3.0
66	Indonesia	3657	Stratovolcano	NaN	2.0
67	Indonesia	3657	Stratovolcano	NaN	2.0
68	Indonesia	3657	Stratovolcano	NaN	2.0
69	Indonesia	3657	Stratovolcano	6.0	1.0
70	Indonesia	3657	Stratovolcano	1.0	2.0
71	Indonesia	3657	Stratovolcano	3.0	3.0
72	Indonesia	3657	Stratovolcano	70.0	3.0
73	Indonesia	3657	Stratovolcano	12.0	3.0
74	Indonesia	3657	Stratovolcano	119.0	2.0
75	Indonesia	3657	Stratovolcano	1.0	3.0
76	Indonesia	2769	Stratovolcano	NaN	2.0
77	Indonesia	1711	Caldera	2.0	2.0

78	Indonesia	2997	Stratovolcano	1028.0	5.0
79	Indonesia	2997	Stratovolcano	120.0	3.0
80	Indonesia	2850	Stratovolcano	60000.0	7.0
81	Indonesia	618	Stratovolcano	2.0	3.0
82	Indonesia	875	Stratovolcano	226.0	3.0
83	Indonesia	875	Stratovolcano	1.0	2.0
84	Indonesia	875	Stratovolcano	NaN	2.0
85	Indonesia	1703	Stratovolcano	2.0	2.0
86	Indonesia	1703	Stratovolcano	1.0	2.0
87	Indonesia	583	Complex volcano	NaN	2.0
88	Indonesia	583	Complex volcano	NaN	1.0
89	Pacific Ocean	728	Stratovolcano	NaN	4.0
90	Pacific Ocean	608	Stratovolcano	NaN	4.0
91	Pacific Ocean	596	Caldera	NaN	3.0
92	Pacific Ocean	596	Caldera	NaN	2.0
93	Pacific Ocean	596	Caldera	4.0	3.0
94	Indonesia	404	Stratovolcano	NaN	4.0
95	Indonesia	1785	Stratovolcano	118.0	2.0
96	Indonesia	1299	Stratovolcano	1.0	2.0
97	Indonesia	725	Stratovolcano	400.0	2.0
98	Indonesia	1797	Stratovolcano	1.0	2.0
99	Indonesia	1797	Stratovolcano	4.0	3.0
100	Indonesia	1797	Stratovolcano	1.0	2.0
101	Indonesia	-5	Submarine volcano	NaN	2.0
102	Indonesia	1318	Stratovolcano	3000.0	3.0
103	Indonesia	1318	Stratovolcano	953.0	3.0
104	Indonesia	1318	Stratovolcano	2806.0	3.0
105	Indonesia	1318	Stratovolcano	1532.0	3.0
106	Indonesia	1318	Stratovolcano	138.0	2.0
107	Indonesia	1318	Stratovolcano	39.0	4.0
108	Indonesia	1635	Stratovolcano	NaN	5.0
109	Indonesia	1714	Stratovolcano	NaN	3.0
110	Indonesia	1714	Stratovolcano	35.0	3.0
111	Indonesia	1714	Stratovolcano	NaN	2.0
112	Indonesia	1714	Stratovolcano	4.0	2.0
113	Indonesia	1714	Stratovolcano	NaN	3.0
114	Indonesia	1714	Stratovolcano	1.0	2.0
115	Indonesia	1714	Stratovolcano	5.0	2.0
116	Indonesia	1357	Stratovolcano	NaN	4.0
117	Indonesia	1357	Stratovolcano	326.0	4.0
118	Indonesia	1357	Stratovolcano	NaN	2.0
119	Philippines	2790	Stratovolcano	NaN	2.0
120	Philippines	1552	Stratovolcano	NaN	2.0
121	Philippines	1552	Stratovolcano	68.0	3.0
122	Philippines	2422	Stratovolcano	3.0	2.0
123	Philippines	1535	Stratovolcano	9.0	2.0
124	Philippines	1535	Stratovolcano	NaN	2.0
125	Philippines	2462	Stratovolcano	49.0	3.0
126	Philippines	2462	Stratovolcano	NaN	2.0
127	Philippines	2462	Stratovolcano	1200.0	4.0
128	Philippines	2462	Stratovolcano	34.0	3.0
129	Philippines	2462	Stratovolcano	3.0	3.0
130	Philippines	2462	Stratovolcano	15.0	3.0
131	Philippines	2462	Stratovolcano	350.0	3.0
132	Philippines	2462	Stratovolcano	NaN	2.0
133	Philippines	2462	Stratovolcano	NaN	2.0

134	Philippines	2462	Stratovolcano	6.0	3.0
135	Philippines	2462	Stratovolcano	1.0	3.0
136	Philippines	311	Stratovolcano	NaN	4.0
137	Philippines	311	Stratovolcano	NaN	3.0
138	Philippines	311	Stratovolcano	12.0	4.0
139	Philippines	311	Stratovolcano	NaN	2.0
140	Philippines	311	Stratovolcano	1335.0	3.0
141	Philippines	311	Stratovolcano	355.0	4.0
142	Philippines	228	Compound volcano	3.0	2.0
143	Taiwan	0	Submarine volcano	NaN	2.0
144	Japan	657	Stratovolcano	8.0	4.0
145	Japan	1117	Stratovolcano	NaN	3.0
146	Japan	1117	Stratovolcano	NaN	5.0
147	Japan	1117	Stratovolcano	153.0	4.0
148	Japan	1117	Stratovolcano	38.0	4.0
149	Japan	1117	Stratovolcano	63.0	4.0
150	Japan	1117	Stratovolcano	NaN	3.0
151	Japan	1700	Shield volcano	NaN	3.0
152	Japan	1700	Shield volcano	1.0	3.0
153	Japan	1700	Shield volcano	1.0	3.0
154	Japan	1483	Complex volcano	15000.0	2.0
155	Japan	1483	Complex volcano	43.0	1.0

	deaths	totalDeaths
1	NaN	16000.0
2	NaN	NaN
3	NaN	NaN
4	NaN	NaN
5	NaN	NaN
6	NaN	NaN
7	NaN	NaN
8	1.0	1.0
9	1.0	1.0
10	1.0	1.0
11	1.0	1.0
12	1.0	1.0
13	2.0	2.0
14	2.0	2.0
15	2.0	2.0
16	3.0	3.0
17	4.0	4.0
18	4.0	4.0
19	4.0	4.0
20	5.0	6.0
21	4.0	4.0
22	5.0	5.0
23	9.0	9.0
24	9.0	9.0
25	2.0	2.0
26	11.0	11.0
27	NaN	NaN
28	20.0	20.0
29	1.0	21.0
30	24.0	24.0
31	25.0	25.0
32	NaN	25.0

33	27.0	27.0
34	37.0	37.0
35	48.0	48.0
36	50.0	50.0
37	59.0	59.0
38	117.0	117.0
39	149.0	149.0
40	3000.0	3000.0
41	100.0	100.0
42	32.0	32.0
43	30.0	30.0
44	200.0	200.0
45	35.0	35.0
46	1369.0	1369.0
47	64.0	64.0
48	6.0	6.0
49	3.0	3.0
50	1.0	1.0
51	66.0	66.0
52	NaN	NaN
53	NaN	NaN
54	22.0	22.0
55	54.0	54.0
56	NaN	NaN
57	NaN	NaN
58	5110.0	5110.0
59	7.0	7.0
60	215.0	215.0
61	35.0	35.0
62	74.0	74.0
63	80.0	80.0
64	221.0	221.0
65	NaN	NaN
66	NaN	NaN
67	NaN	NaN
68	NaN	NaN
69	6.0	6.0
70	1.0	1.0
71	3.0	3.0
72	70.0	70.0
73	12.0	12.0
74	119.0	119.0
75	1.0	1.0
76	NaN	NaN
77	2.0	2.0
78	1028.0	1028.0
79	120.0	120.0
80	11000.0	60000.0
81	2.0	2.0
82	98.0	226.0
83	1.0	1.0
84	NaN	NaN
85	2.0	2.0
86	1.0	1.0
87	NaN	NaN
88	NaN	NaN

89	NaN	NaN
90	NaN	NaN
91	NaN	NaN
92	NaN	NaN
93	4.0	4.0
94	NaN	NaN
95	NaN	118.0
96	1.0	1.0
97	NaN	400.0
98	1.0	1.0
99	4.0	4.0
100	1.0	1.0
101	NaN	NaN
102	3000.0	3000.0
103	953.0	953.0
104	2806.0	2806.0
105	1532.0	1532.0
106	NaN	138.0
107	39.0	39.0
108	NaN	NaN
109	NaN	NaN
110	35.0	35.0
111	NaN	NaN
112	4.0	4.0
113	NaN	NaN
114	1.0	1.0
115	5.0	5.0
116	NaN	NaN
117	326.0	326.0
118	NaN	NaN
119	NaN	NaN
120	NaN	NaN
121	68.0	68.0
122	3.0	3.0
123	NaN	9.0
124	NaN	NaN
125	49.0	49.0
126	NaN	NaN
127	1200.0	1200.0
128	34.0	34.0
129	3.0	3.0
130	15.0	15.0
131	350.0	350.0
132	NaN	NaN
133	NaN	NaN
134	6.0	6.0
135	1.0	1.0
136	NaN	NaN
137	NaN	NaN
138	NaN	12.0
139	NaN	NaN
140	1335.0	1335.0
141	355.0	355.0
142	NaN	3.0
143	NaN	NaN
144	8.0	8.0

145	NaN	NaN
146	NaN	NaN
147	153.0	153.0
148	23.0	38.0
149	28.0	63.0
150	NaN	NaN
151	NaN	NaN
152	1.0	1.0
153	1.0	1.0
154	NaN	15000.0
155	43.0	43.0

TASK 1.3: Tailoring Time (10 Points)

It's pretty obvious that the year, month, and day look pretty weird in the dataset. We're going to have to do some hardcore cleaning on the [time](#).

We need to have only ONE column called "date" that contains the full date (YYYY-MM-DD), not separated into three columns.

Make sure there are no floating point values in the date and sort the data from most recent to least.

Remove the old columns and place the new column next to the 'id' column.

YOU CAN USE MULTIPLE LINES OF CODE BUT CAN'T USE LOOPS.

Note: It is alright to have only a **maximum of 12 NaTs** for some dates that often go further back than the 1600s because the datetime module in Pandas has a limit (unless otherwise guided).

```
In [38]: # Drop original columns and sort.
new_df['date'] = pd.to_datetime(new_df[['year', 'month', 'day']], errors='coerce')
new_df = new_df.sort_values(by='date', ascending=False)
new_df = new_df.drop(columns=['year', 'month', 'day'])

# Place new column next to 'id' column.
cols_names = new_df.columns.tolist()
cols_names.insert(1, cols_names.pop(cols_names.index('date')))
new_df = new_df[cols_names]

# KEEP THIS. It will display the whole dataframe.
new_df
```

Out[38]:

	id	date	tsunamiEventId	earthquakeEventId	volcanoLocationId	volcanoLocationName
122	693	1996-08-10	NaN	NaN	70202	
51	112	1994-11-22	NaN	NaN	60325	
155	933	1991-06-03	NaN	NaN	80210	
61	144	1990-02-10	NaN	NaN	60328	
93	486	1988-05-09	NaN	NaN	60509	
25	31	1987-04-17	NaN	NaN	10106	
50	111	1986-12-31	NaN	NaN	60325	
9	13	1986-07-24	NaN	NaN	10104	
72	203	1985-05-10	NaN	NaN	60330	
135	739	1984-09-09	NaN	NaN	70303	
34	40	1984-08-15	5807.0	NaN	20403	
88	447	1983-08-17	4702.0	NaN	60425	
94	488	1983-07-23	NaN	NaN	60601	
84	398	1981-09-05	NaN	NaN	60415	
75	206	1981-03-29	NaN	NaN	60330	
24	30	1979-09-12	NaN	NaN	10106	
15	20	1979-03-08	NaN	NaN	50103	
39	49	1979-02-20	NaN	NaN	60320	
73	204	1978-09-19	NaN	NaN	60330	

	id	date	tsunamiEventId	earthquakeEventId	volcanoLocationId	volcanoLocationName
124	698	1978-07-29	NaN	NaN	70301	
5	7	1977-01-10	NaN	NaN	20303	
100	562	1976-09-15	NaN	NaN	60702	
74	205	1976-08-31	NaN	NaN	60330	
99	561	1974-02-11	NaN	NaN	60702	
16	21	1970-03-02	NaN	NaN	10101	
142	777	1969-03-21	2872.0	NaN	70402	
81	389	1969-01-28	NaN	NaN	60411	
49	105	1969-01-07	NaN	NaN	60325	
134	736	1968-04-21	NaN	NaN	70303	
71	202	1967-08-31	NaN	NaN	60330	
107	589	1966-08-12	NaN	NaN	60704	
60	141	1966-04-26	NaN	NaN	60328	
141	762	1965-09-28	1977.0	NaN	70307	
83	395	1964-01-01	NaN	NaN	60415	
77	345	1963-09-05	NaN	NaN	60401	
79	353	1963-05-16	NaN	4292.0	60402	
70	201	1963-05-05	NaN	NaN	60330	
78	351	1963-03-18	1942.0	NaN	60402	

	id	date	tsunamiEventId	earthquakeEventId	volcanoLocationId	volcanoLocationName
115	657	1962-12-31	NaN	NaN	60806	
48	104	1961-05-08	NaN	NaN	60325	
96	525	1958-07-12	NaN	NaN	60611	
150	878	1955-10-13	NaN	NaN	80208	
32	38	1954-08-03	NaN	NaN	50101	
47	100	1954-01-18	NaN	NaN	60325	
59	140	1951-08-31	NaN	NaN	60328	
121	680	1950-09-15	NaN	NaN	70108	
69	200	1950-08-28	NaN	NaN	60330	
87	441	1948-04-07	NaN	NaN	60425	
133	735	1947-01-07	NaN	NaN	70303	
68	199	1946-10-29	NaN	NaN	60330	
38	45	1944-12-04	NaN	NaN	60320	
33	39	1944-03-27	NaN	NaN	10102	
67	196	1941-09-21	NaN	NaN	60330	
98	552	1940-06-20	NaN	NaN	60702	
132	733	1938-06-05	NaN	NaN	70303	
123	697	1933-12-25	1666.0	NaN	70301	
144	823	1933-12-24	NaN	NaN	80205	

	id	date	tsunamiEventId	earthquakeEventId	volcanoLocationId	volcanoLocationName
46	94	1930-12-18	NaN	NaN	60325	
20	26	1930-09-11	4218.0	NaN	10104	
22	28	1928-11-08	NaN	NaN	10106	
82	394	1928-08-04	1607.0	NaN	60415	
35	42	1926-01-23	NaN	NaN	10204	
45	92	1920-07-25	NaN	NaN	60325	
19	25	1919-05-22	1476.0	NaN	10104	
58	138	1919-05-19	NaN	NaN	60328	
14	18	1917-04-01	NaN	NaN	40105	
26	32	1914-09-10	NaN	NaN	40104	
149	867	1914-01-12	1412.0	2994.0	80208	
66	195	1913-06-23	NaN	NaN	60330	
106	585	1913-03-14	1407.0	2973.0	60704	
28	34	1912-12-03	NaN	NaN	20302	
65	193	1911-11-08	NaN	NaN	60330	
140	761	1911-01-30	1392.0	8391.0	70307	
64	191	1909-08-29	NaN	NaN	60330	
86	411	1907-09-28	NaN	NaN	60418	
7	11	1907-08-04	NaN	NaN	201112	

	id	date	tsunamiEventId	earthquakeEventId	volcanoLocationId	volcanoLocationName
8	12	1905-03-10	NaN	NaN	10102	
11	15	1904-02-25	NaN	NaN	30301	
21	27	1903-08-30	NaN	NaN	40105	
57	137	1901-05-22	NaN	NaN	60328	
131	730	1897-05-23	NaN	NaN	70303	
29	35	1895-06-17	NaN	NaN	50103	
63	176	1895-05-22	NaN	NaN	60330	
105	584	1892-06-07	1197.0	NaN	60704	
118	669	1890-06-29	NaN	NaN	60807	
101	569	1889-09-06	1181.0	6309.0	60703	
3	4	1888-03-13	1175.0	NaN	50107	
130	724	1887-03-09	NaN	NaN	70303	
27	33	1886-08-31	NaN	NaN	40311	
62	167	1885-04-18	NaN	NaN	60330	
12	16	1878-02-04	1095.0	6195.0	50214	
56	136	1875-01-29	NaN	NaN	60328	
139	756	1874-07-19	NaN	NaN	70307	
119	677	1873-01-16	NaN	NaN	70106	

	id	date	tsunamiEventId	earthquakeEventId	volcanoLocationId	volcanoLocationName
44	82	1872-11-03	NaN	NaN	60325	
23	29	1872-04-24	NaN	NaN	10102	
43	81	1872-04-15	NaN	NaN	60325	
129	716	1871-12-08	NaN	NaN	70303	
114	643	1871-08-07	NaN	NaN	60806	
120	679	1871-04-30	1020.0	2179.0	70108	
97	533	1871-03-03	1022.0	NaN	60701	
85	410	1869-07-07	NaN	NaN	60418	
55	135	1864-01-04	NaN	NaN	60328	
117	666	1861-12-29	NaN	NaN	60807	
104	581	1856-03-02	859.0	1995.0	60704	
143	787	1853-10-29	809.0	NaN	80103	
128	712	1853-07-13	NaN	NaN	70303	
31	37	1853-06-24	NaN	NaN	40311	
54	133	1848-05-16	NaN	NaN	60328	
95	489	1845-02-08	741.0	1865.0	60603	
37	44	1843-11-25	NaN	NaN	10106	
113	629	1840-02-02	718.0	NaN	60806	
112	627	1838-02-26	NaN	NaN	60806	

	id	date	tsunamiEventId	earthquakeEventId	volcanoLocationId	volcanoLocationName
42	71	1832-12-25	NaN	NaN	60325	
4	6	1832-11-01	NaN	NaN	10106	
53	132	1826-10-13	NaN	NaN	60328	
41	69	1822-12-27	NaN	NaN	60325	
92	482	1820-06-11	NaN	NaN	60509	
76	333	1817-01-24	NaN	NaN	60335	
80	367	1815-04-10	613.0	NaN	60404	
127	707	1814-02-01	NaN	NaN	70303	
103	580	1812-08-06	NaN	NaN	60704	
18	24	1805-08-11	NaN	NaN	10102	
126	706	1800-10-30	NaN	NaN	70303	
154	932	1792-05-21	568.0	1515.0	80210	
148	859	1781-04-11	529.0	NaN	80208	
147	858	1779-11-08	NaN	NaN	80208	
6	9	1779-08-08	NaN	NaN	10102	
111	618	1773-02-02	NaN	NaN	60806	
110	617	1772-05-09	504.0	NaN	60806	
125	704	1766-07-20	NaN	NaN	70303	
138	754	1754-11-28	2853.0	1328.0	70307	

	id	date	tsunamiEventId	earthquakeEventId	volcanoLocationId	volcanoLocationName
137	753	1749-08-11	431.0	1304.0	70307	
13	17	1737-05-20	NaN	NaN	10102	
153	916	1717-02-07	NaN	NaN	80209	
152	915	1716-11-09	NaN	NaN	80209	
136	752	1716-09-24	387.0	1205.0	70307	
52	128	1716-07-20	NaN	NaN	60328	
102	578	1711-12-11	NaN	NaN	60704	
91	477	1694-11-20	NaN	NaN	60509	
90	466	1692-06-04	NaN	NaN	60507	
17	23	1682-08-12	NaN	NaN	10102	
1	1	NaT	2852.0	421.0	10106	
2	2	NaT	NaN	NaN	10106	
10	14	NaT	NaN	NaN	10106	
30	36	NaT	NaN	NaN	10101	
36	43	NaT	NaN	920.0	201141	
40	64	NaT	NaN	NaN	60325	
89	457	NaT	285.0	NaN	60505	
108	599	NaT	312.0	NaN	60804	
109	607	NaT	222.0	NaN	60806	
116	664	NaT	5754.0	10519.0	60807	
145	844	NaT	61.0	7341.0	80208	
146	846	NaT	NaN	NaN	80208	
151	904	NaT	NaN	NaN	80209	

Part 2: Volcanic Matryoshkas 🧩 (30 POINTS TOTAL)

Now, that most of the data has been tidied up. We will organize the data into more sizable pieces of information in order to extract useful information.

2.1.1: (10 points here)

Use the **groupby function in Pandas** to create separate dataframes for each unique country.

- **Each table must only have the columns:** 'date' 'country', 'name', and 'vei' (Having the index is ok. Up to you.)
- **Sort** the dataframe of **each country** by **highest to lowest 'vei'**
- Use the **display** function to show **each sorted table**

You MUST use the groupby function here and display your results.

```
In [39]: from IPython.display import display

for country, groupings in new_df.groupby(by='country'):
    unique_country = groupings[['date', 'country', 'name', 'vei']]
    unique_country = unique_country.sort_values(by='vei', ascending=False)
    display(unique_country)
```

	date	country	name	vei
34	1984-08-15	Cameroon	Oku Volcanic Field	NaN

	date	country	name	vei
11	1904-02-25	Comoros	Karthala	2.0

	date	country	name	vei
28	1912-12-03	Congo, DRC	Nyamulagira	3.0
5	1977-01-10	Congo, DRC	Nyiragongo	1.0

	date	country	name	vei
7	1907-08-04	Ethiopia	Alayta	2.0
36	NaT	Ethiopia	Dama Ali	NaN

	date	country	name	vei
35	1926-01-23	Greece	Santorini	2.0

	date	country	name	vei
80	1815-04-10	Indonesia	Tambora	7.0
78	1963-03-18	Indonesia	Agung	5.0
108	NaT	Indonesia	Gamkonora	5.0
59	1951-08-31	Indonesia	Kelud	4.0
107	1966-08-12	Indonesia	Awu	4.0
44	1872-11-03	Indonesia	Merapi	4.0
58	1919-05-19	Indonesia	Kelud	4.0
60	1966-04-26	Indonesia	Kelud	4.0
61	1990-02-10	Indonesia	Kelud	4.0
94	1983-07-23	Indonesia	Colo	4.0
43	1872-04-15	Indonesia	Merapi	4.0
116	NaT	Indonesia	Kie Besi	4.0
117	1861-12-29	Indonesia	Kie Besi	4.0
53	1826-10-13	Indonesia	Kelud	4.0
40	NaT	Indonesia	Merapi	3.0
103	1812-08-06	Indonesia	Awu	3.0
75	1981-03-29	Indonesia	Semeru	3.0
72	1985-05-10	Indonesia	Semeru	3.0
73	1978-09-19	Indonesia	Semeru	3.0
51	1994-11-22	Indonesia	Merapi	3.0
105	1892-06-07	Indonesia	Awu	3.0
54	1848-05-16	Indonesia	Kelud	3.0
104	1856-03-02	Indonesia	Awu	3.0
46	1930-12-18	Indonesia	Merapi	3.0
48	1961-05-08	Indonesia	Merapi	3.0
71	1967-08-31	Indonesia	Semeru	3.0
99	1974-02-11	Indonesia	Karangetang	3.0
57	1901-05-22	Indonesia	Kelud	3.0
82	1928-08-04	Indonesia	Paluweh	3.0
79	1963-05-16	Indonesia	Agung	3.0

	date	country	name	vei
65	1911-11-08	Indonesia	Semeru	3.0
81	1969-01-28	Indonesia	Iya	3.0
110	1772-05-09	Indonesia	Gamalama	3.0
41	1822-12-27	Indonesia	Merapi	3.0
113	1840-02-02	Indonesia	Gamalama	3.0
109	NaT	Indonesia	Gamalama	3.0
102	1711-12-11	Indonesia	Awu	3.0
49	1969-01-07	Indonesia	Merapi	2.0
100	1976-09-15	Indonesia	Karangetang	2.0
50	1986-12-31	Indonesia	Merapi	2.0
86	1907-09-28	Indonesia	Lewotobi	2.0
84	1981-09-05	Indonesia	Paluweh	2.0
83	1964-01-01	Indonesia	Paluweh	2.0
118	1890-06-29	Indonesia	Kie Besi	2.0
74	1976-08-31	Indonesia	Semeru	2.0
106	1913-03-14	Indonesia	Awu	2.0
64	1909-08-29	Indonesia	Semeru	2.0
45	1920-07-25	Indonesia	Merapi	2.0
38	1944-12-04	Indonesia	Dieng Volcanic Complex	2.0
98	1940-06-20	Indonesia	Karangetang	2.0
66	1913-06-23	Indonesia	Semeru	2.0
87	1948-04-07	Indonesia	Iliwerung	2.0
68	1946-10-29	Indonesia	Semeru	2.0
67	1941-09-21	Indonesia	Semeru	2.0
77	1963-09-05	Indonesia	Batur	2.0
70	1963-05-05	Indonesia	Semeru	2.0
115	1962-12-31	Indonesia	Gamalama	2.0
96	1958-07-12	Indonesia	Mahawu	2.0
47	1954-01-18	Indonesia	Merapi	2.0
76	1817-01-24	Indonesia	Ijen	2.0

	date	country	name	vei
97	1871-03-03	Indonesia	Ruang	2.0
95	1845-02-08	Indonesia	Soputan	2.0
55	1864-01-04	Indonesia	Kelud	2.0
114	1871-08-07	Indonesia	Gamalama	2.0
63	1895-05-22	Indonesia	Semeru	2.0
62	1885-04-18	Indonesia	Semeru	2.0
101	1889-09-06	Indonesia	Banua Wuhu	2.0
52	1716-07-20	Indonesia	Kelud	2.0
42	1832-12-25	Indonesia	Merapi	2.0
85	1869-07-07	Indonesia	Lewotobi	2.0
112	1838-02-26	Indonesia	Gamalama	2.0
111	1773-02-02	Indonesia	Gamalama	2.0
88	1983-08-17	Indonesia	Iliwerung	1.0
39	1979-02-20	Indonesia	Dieng Volcanic Complex	1.0
69	1950-08-28	Indonesia	Semeru	1.0
56	1875-01-29	Indonesia	Kelud	NaN

	date	country	name	vei
33	1944-03-27	Italy	Vesuvius	3.0
24	1979-09-12	Italy	Etna	3.0
10	NaT	Italy	Etna	3.0
30	NaT	Italy	Campi Flegrei	3.0
20	1930-09-11	Italy	Stromboli	3.0
19	1919-05-22	Italy	Stromboli	3.0
23	1872-04-24	Italy	Vesuvius	3.0
2	NaT	Italy	Etna	3.0
6	1779-08-08	Italy	Vesuvius	3.0
13	1737-05-20	Italy	Vesuvius	3.0
17	1682-08-12	Italy	Vesuvius	3.0
25	1987-04-17	Italy	Etna	2.0
9	1986-07-24	Italy	Stromboli	2.0
8	1905-03-10	Italy	Vesuvius	2.0
4	1832-11-01	Italy	Etna	2.0
37	1843-11-25	Italy	Etna	2.0
18	1805-08-11	Italy	Vesuvius	2.0
22	1928-11-08	Italy	Etna	1.0
16	1970-03-02	Italy	Campi Flegrei	NaN
1	NaT	Italy	Etna	NaN

	date	country	name	vei
146	NaT	Japan	Aira	5.0
149	1914-01-12	Japan	Aira	4.0
148	1781-04-11	Japan	Aira	4.0
144	1933-12-24	Japan	Kuchinoerabujima	4.0
147	1779-11-08	Japan	Aira	4.0
150	1955-10-13	Japan	Aira	3.0
145	NaT	Japan	Aira	3.0
153	1717-02-07	Japan	Kirishimayama	3.0
151	NaT	Japan	Kirishimayama	3.0
152	1716-11-09	Japan	Kirishimayama	3.0
154	1792-05-21	Japan	Unzendake	2.0
155	1991-06-03	Japan	Unzendake	1.0

	date	country	name	vei
14	1917-04-01	New Zealand	Okataina	1.0
21	1903-08-30	New Zealand	Okataina	1.0
26	1914-09-10	New Zealand	Whakaari/White Island	NaN

	date	country	name	vei
89	NaT	Pacific Ocean	Teon	4.0
90	1692-06-04	Pacific Ocean	Serua	4.0
93	1988-05-09	Pacific Ocean	Banda Api	3.0
91	1694-11-20	Pacific Ocean	Banda Api	3.0
92	1820-06-11	Pacific Ocean	Banda Api	2.0

	date	country	name	vei
12	1878-02-04	Papua New Guinea	Rabaul	3.0
15	1979-03-08	Papua New Guinea	Karkar	2.0
32	1954-08-03	Papua New Guinea	Bam	2.0
29	1895-06-17	Papua New Guinea	Karkar	2.0
3	1888-03-13	Papua New Guinea	Ritter Island	2.0

	date	country	name	vei
141	1965-09-28	Philippines	Taal	4.0
127	1814-02-01	Philippines	Mayon	4.0
136	1716-09-24	Philippines	Taal	4.0
138	1754-11-28	Philippines	Taal	4.0
131	1897-05-23	Philippines	Mayon	3.0
135	1984-09-09	Philippines	Mayon	3.0
134	1968-04-21	Philippines	Mayon	3.0
121	1950-09-15	Philippines	Camiguin	3.0
125	1766-07-20	Philippines	Mayon	3.0
137	1749-08-11	Philippines	Taal	3.0
129	1871-12-08	Philippines	Mayon	3.0
130	1887-03-09	Philippines	Mayon	3.0
128	1853-07-13	Philippines	Mayon	3.0
140	1911-01-30	Philippines	Taal	3.0
124	1978-07-29	Philippines	Bulusan	2.0
142	1969-03-21	Philippines	Didicas	2.0
133	1947-01-07	Philippines	Mayon	2.0
122	1996-08-10	Philippines	Kanlaon	2.0
123	1933-12-25	Philippines	Bulusan	2.0
132	1938-06-05	Philippines	Mayon	2.0
119	1873-01-16	Philippines	Ragang	2.0
139	1874-07-19	Philippines	Taal	2.0
126	1800-10-30	Philippines	Mayon	2.0
120	1871-04-30	Philippines	Camiguin	2.0

	date	country	name	vei
143	1853-10-29	Taiwan	Unnamed	2.0

	date	country	name	vei
27	1886-08-31	Tonga	Niuafo'ou	4.0
31	1853-06-24	Tonga	Niuafo'ou	0.0

2.1.2: (5 points here)

Using **groupby** again, print out the **maximum 'vei'** for **each unique country**.

You MUST use the groupby function here and print your results.

- **Print** out your results in a format like the following: "Country: {country_name}, Highest VEI: {vei}"

```
In [40]: max_vei = new_df.groupby('country')['vei'].max()

for country, vei_max in max_vei.items():
    print(f"Country: {country}, Highest Vei: {vei_max}")
```

```
Country: Cameroon, Highest Vei: nan
Country: Comoros, Highest Vei: 2.0
Country: Congo, DRC, Highest Vei: 3.0
Country: Ethiopia, Highest Vei: 2.0
Country: Greece, Highest Vei: 2.0
Country: Indonesia, Highest Vei: 7.0
Country: Italy, Highest Vei: 3.0
Country: Japan, Highest Vei: 5.0
Country: New Zealand, Highest Vei: 1.0
Country: Pacific Ocean, Highest Vei: 4.0
Country: Papua New Guinea, Highest Vei: 3.0
Country: Philippines, Highest Vei: 4.0
Country: Taiwan, Highest Vei: 2.0
Country: Tonga, Highest Vei: 4.0
```

Finally, we have ALMOST REACHED THE END!! Since there is still quite a bit of missing data, we want to make use of what is still available.

A very powerful tool in Python's magnificent collection of libraries is its beautiful graphing tools.

Check out libraries such as [Seaborn](#) or [Matplotlib](#) to create meaningful visualizations! **Your final task in this section requires the use of these libraries**

2.1.3: (15 points here)

- Based on the **unique names of volcanoes**, filter names that have more than 4 **datapoints under their name**.
- Each datapoint in the dataframe refers to a recorded instance of a volcanic eruption.
- Make **separate line graphs for each volcano** and **plot their VEIs over time**.

Make sure to properly label all parts of the graph appropriately to receive credit 🙋
(like title, axes, legend, etc...)

```
In [41]: # REMOVE THIS COMMENT AND ANSWER STARTING HERE
# not adding the graphs here, but the graphs should have the volcano name as the ti
# the x axis should be the years and the y axis should be the VEI
```

```

# A graph that doesn't have these things will only get partial points

import matplotlib.pyplot as plt
import seaborn as sns

volcano_counts = new_df['name'].value_counts()
volcanoes_to_plot = volcano_counts[volcano_counts > 4].index.tolist()

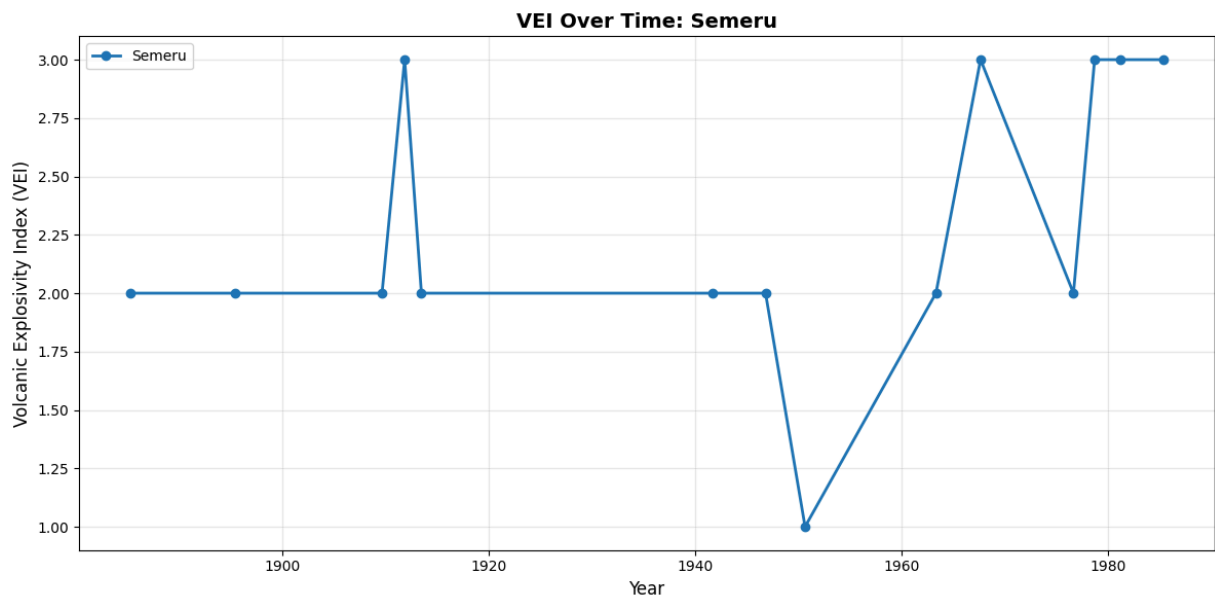
for volcano in volcanoes_to_plot:
    volcano_data = new_df[new_df['name'] == volcano].copy()
    volcano_data = volcano_data.sort_values('date')

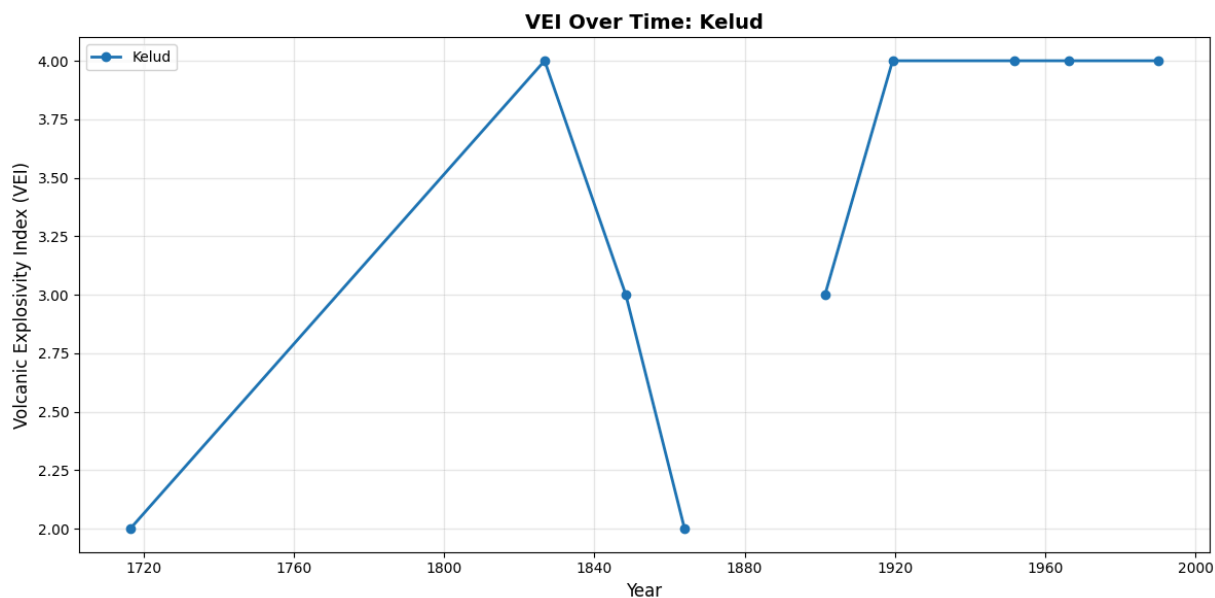
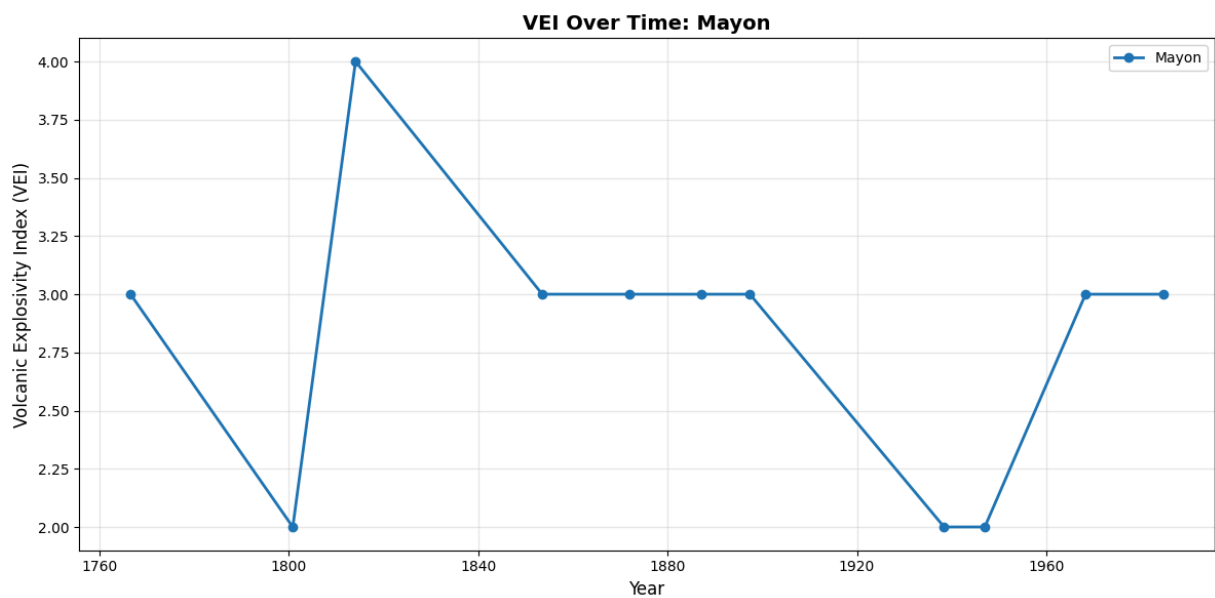
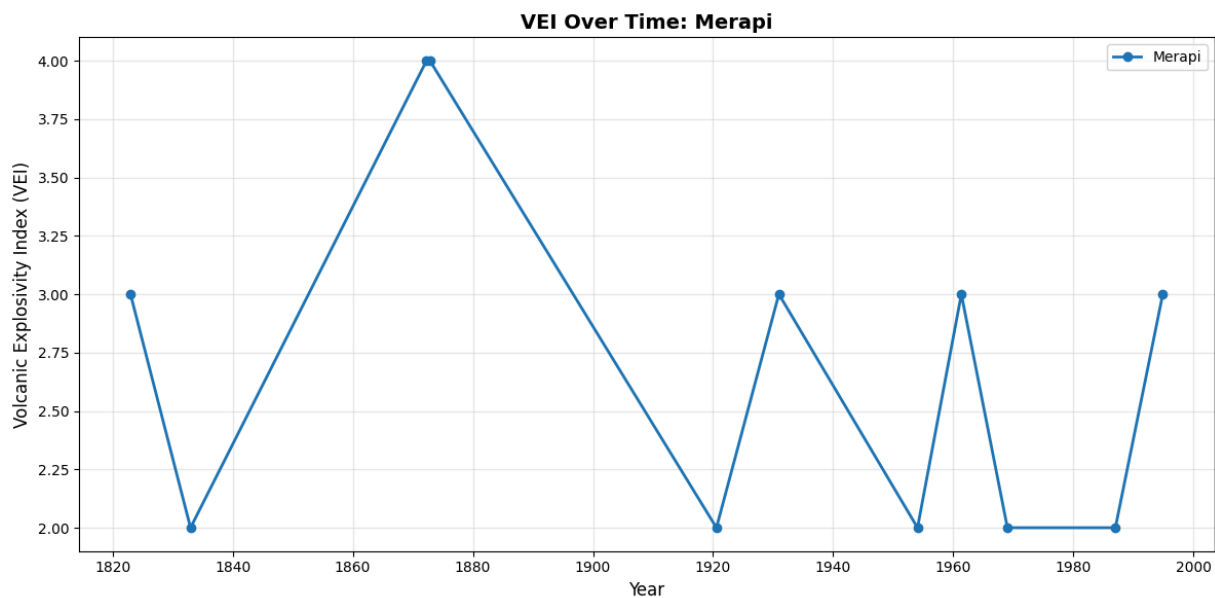
    plt.figure(figsize=(12, 6))
    plt.plot(volcano_data['date'], volcano_data['vei'], marker='o', linewidth=2, ma

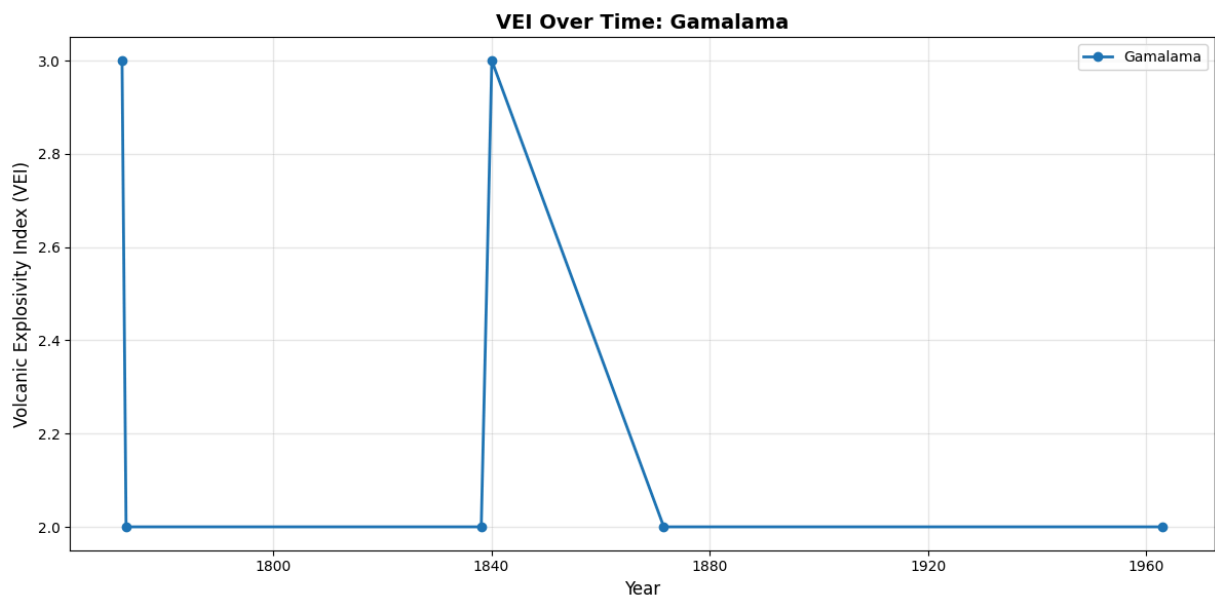
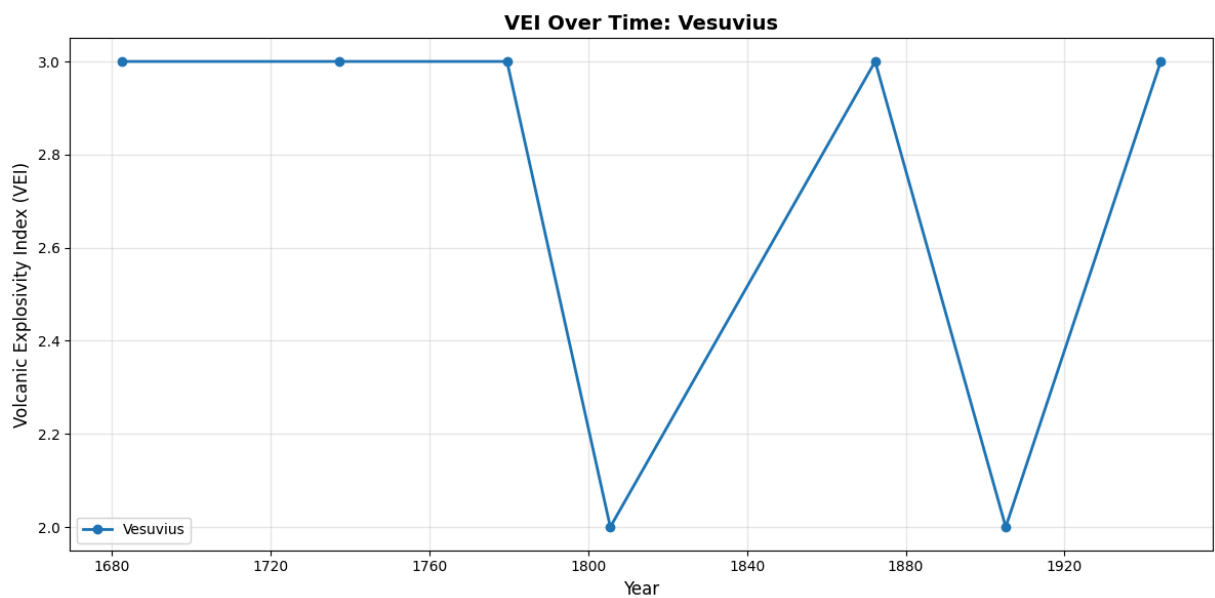
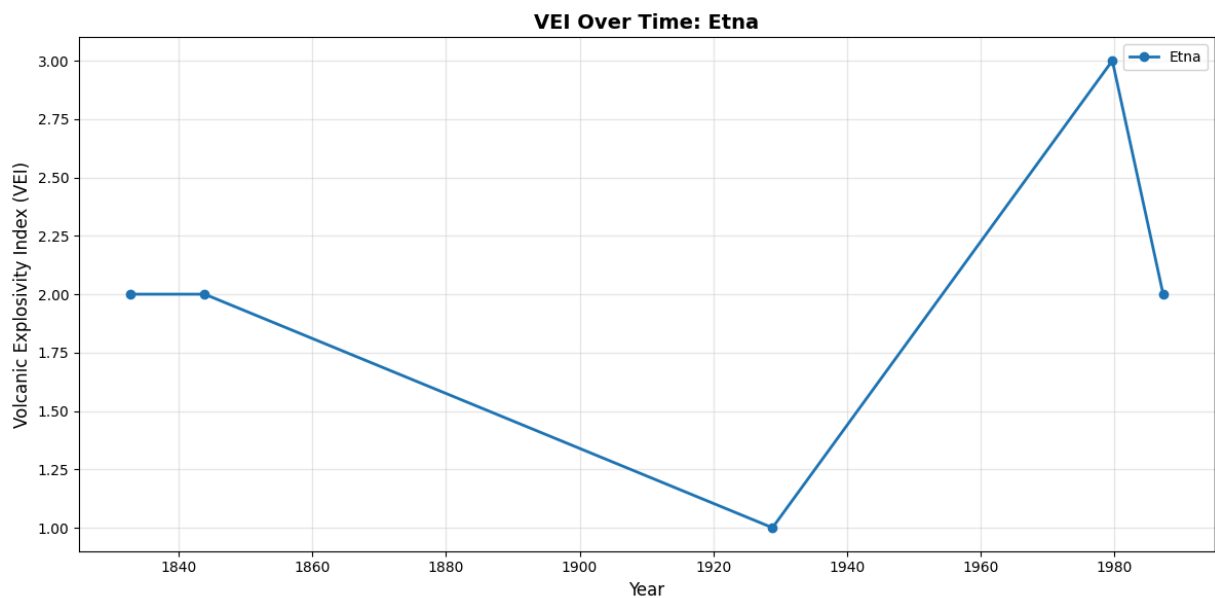
    plt.title(f'VEI Over Time: {volcano}', fontsize=14, fontweight='bold')
    plt.xlabel('Year', fontsize=12)
    plt.ylabel('Volcanic Explosivity Index (VEI)', fontsize=12)

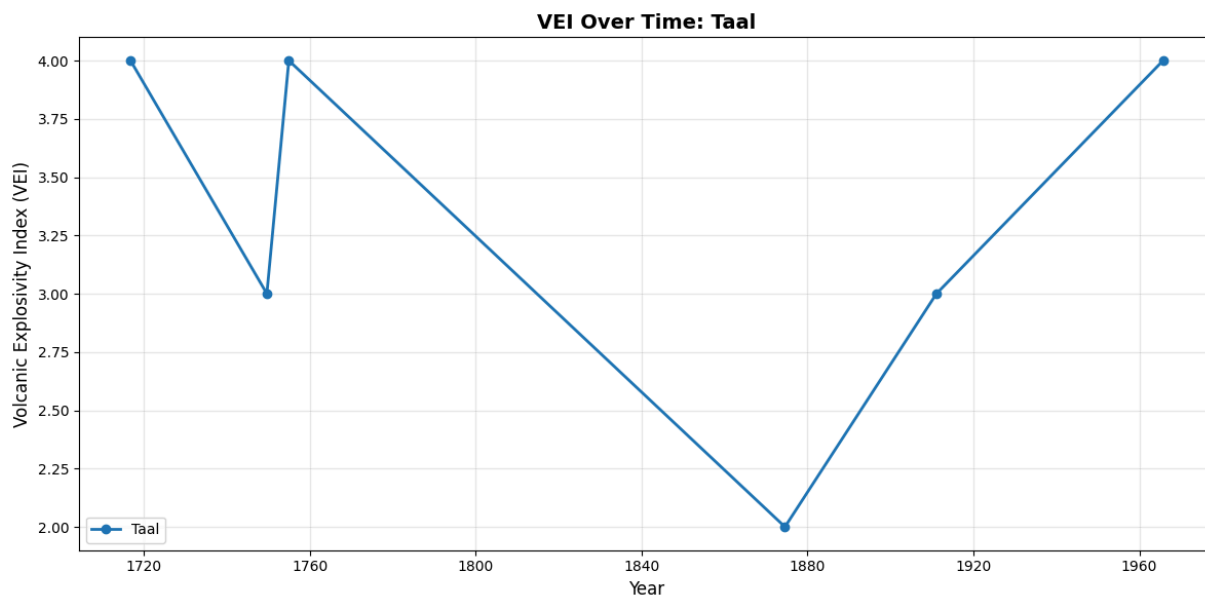
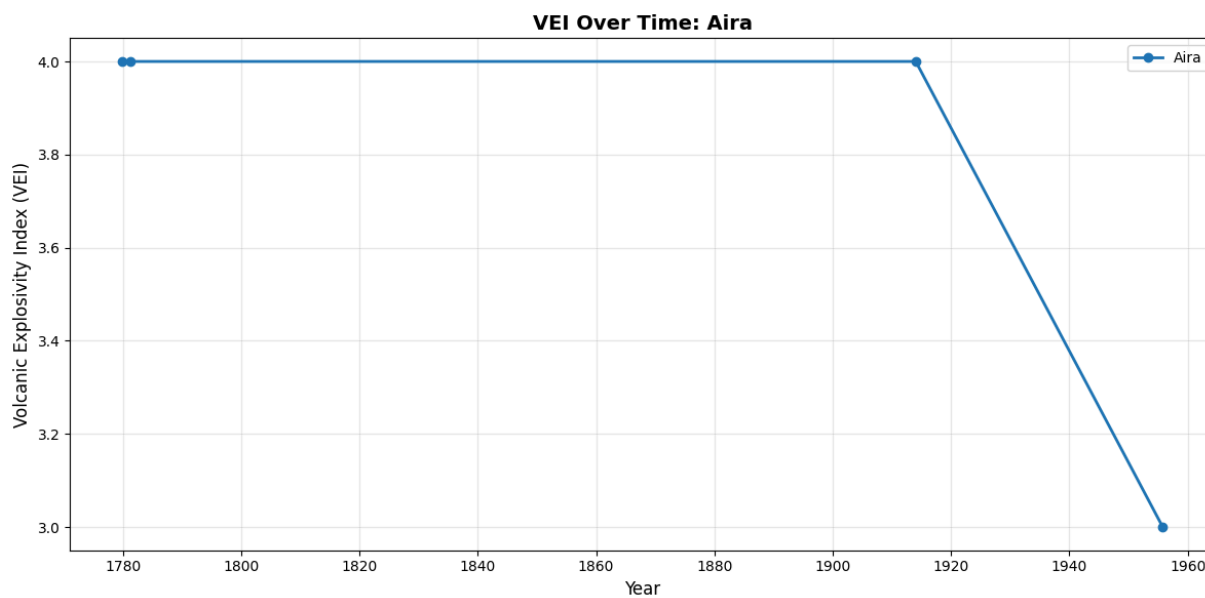
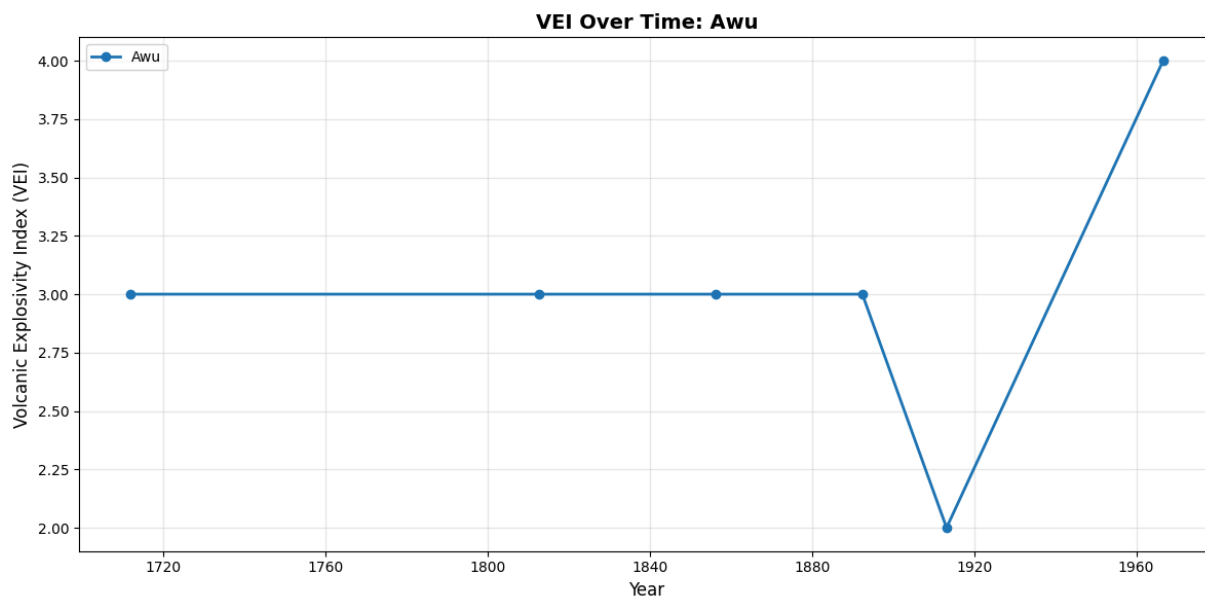
    plt.grid(True, alpha=0.3)
    plt.legend()
    plt.tight_layout()
    plt.show()

```









Part 3: Fiery Jobs 🚒 (15 POINTS TOTAL)

Proficiency in SQL is also super important. SQL databases are essentially relational databases in which there are vast amounts of tabular data, which can often be used to connect with related tabular data. [This](#) is a pretty good intro into learning more about SQL.

Check out this [tutorial](#) for some clarifications on SQL.

Now! We'll be using `sqlite` to access a database.

- **!!!IMPORTANT!!! PLEASE Start by downloading the sql lite file and putting it in the same directory as this [KAGGLE DATASET URL](#) (hit the 'download' button in the upper right). You need an account in order to download the dataset.**
- Check out the description of the data so you know the table / column names.

The following code will use `sqlite3` to create a database connection. `sqlite3` is the library in Python that assists in navigating through SQL databases.

Note: If you are working on this assignment via Google Colab, sometimes the runtime resets and it will throw errors.

Instead of running through the entire notebook, run the notebook from the following code block and onwards:

- Click anywhere on the next code block.
- Go up to where it says '**Runtime**' in the toolbar (right under the title of the notebook and **in between 'Insert' and 'Tools'**)
- Hover over it and **click on the option** that says '**Run cell and below**'

```
In [42]: import sqlite3
import pandas as pd # Pandas was already imported from the previous sections

conn = sqlite3.connect("database.sqlite")
crsr = conn.cursor()
```

!!!IMPORTANT!!! PLEASE Start by downloading the sql lite file and putting it in the same directory as this [KAGGLE DATASET URL](#) (hit the 'download' button in the upper right). You need an account in order to download the dataset.

If you cannot see anything when executing this code, that means you did not download the Kaggle dataset correctly.

```
In [43]: # This code will let you check out the different tables within the database.
query = "SELECT name FROM sqlite_master WHERE type='table';"
```

```
tables = crsr.execute(query).fetchall()
print(tables)
```

```
[('Salaries',), ('Salaries_2010',), ('Salaries_2011',), ('Salaries_2012',), ('Salaries_2013',), ('Salaries_2014',), ('Salaries_2015',)]
```

Remember that each problem should be solved with a single SQL query.

Note: All outputs must be shown

- Only include whatever fields are mentioned throughout each question, nothing more and nothing less.
- Follow each instruction clearly

3.1.1: 2 Points

From the **Salaries** table, get the **average base pay** for firefighters (all job titles consisting of the word "firefighter" (**not case-sensitive**)) between the **years 2010 to 2015**.

Remember that firefighters that also occupy other professions are still considered firefighters.

Hint: Look into [this](#) 🐞

```
In [44]: query = "SELECT AVG(BasePay) AS AverageBasePay " \
"FROM Salaries " \
"WHERE year BETWEEN 2010 AND 2015 " \
"AND JobTitle LIKE '%firefighter%'"

# KEEP THIS. It will display the whole dataframe.
df = pd.read_sql(query, conn)
df
```

```
Out[44]:
```

	AverageBasePay
0	101199.486996

3.1.2: 2 Points

From the **Salaries** table, get all the firefighters (all job titles consisting of the word "firefighter" (**not case-sensitive**)) in the **year 2011** making under **\$40,000 as a base pay**. **Sort** them in **descending** order by their pay.

Remember that firefighters that also occupy other professions are still considered firefighters.

```
In [45]: query = "SELECT EmployeeName, JobTitle, BasePay " \
"FROM Salaries " \
"WHERE JobTitle LIKE '%firefighter%' " \
"AND year = 2011 " \
"AND BasePay < 40000 ORDER BY BasePay DESC"

# KEEP THIS. It will display the whole dataframe.
```

```
df = pd.read_sql(query, conn)
df
```

Out[45]:

	EmployeeName	JobTitle	BasePay
0	MEIBEK SCOTT-CHUNG	FIREFIGHTER	37495.09
1	LAURA ATTIA	FIREFIGHTER	35779.10
2	MICHAEL CUDDY	FIREFIGHTER	32058.01
3	FELIX DUAG	FIREFIGHTER	20658.44
4	DANIEL ARCINIEGA	FIREFIGHTER	13345.28
5	GREG BARRON	FIREFIGHTER	13046.35
6	JAMES MOORE	FIREFIGHTER	7896.00
7	ALLISON KALHAMMER	FIREFIGHTER	7477.60
8	DARREN MILLER	FIREFIGHTER	7477.60
9	MARTIN GREALISH	FIREFIGHTER	6880.50
10	SARAH HAMILTON	FIREFIGHTER	4350.41
11	ANGELA FREEMAN	FIREFIGHTER	0.00
12	WILLIAM KNEIS	FIREFIGHTER	0.00
13	MARK CHERRINGTON	FIREFIGHTER	0.00
14	JOHN BENSON	FIREFIGHTER	0.00
15	ERIKA LOCKHART	FIREFIGHTER	0.00
16	CHARLES JOHNSON	FIREFIGHTER	0.00
17	ROBERTO LUCHA	FIREFIGHTER	0.00
18	ROCK CRAWFORD	FIREFIGHTER	0.00

3.1.3: 4 Points

From the **Salaries** table, first get the **averages** of **base pay**, **benefits**, and **overtime pay** for firefighters (all job titles consisting of the word "firefighter" (**not case-sensitive**)).

- Then, make a **column with the sum** of these **three averages**
- Finally, **exclude** job titles containing "FIREFIGHTER" (**case-sensitive**)

Remember that firefighters that also occupy other professions are still considered firefighters.

```
In [46]: query = "SELECT AVG(BasePay) AS AverageBasePay, " \
"AVG(Benefits) AS AverageBenefits, " \
"AVG(OvertimePay) AS AverageOvertimePay, " \
"(AVG(BasePay)+AVG(Benefits)+AVG(OvertimePay)) AS SumOfAverages " \
"FROM Salaries " \
```

```
"WHERE JobTitle LIKE '%firefighter%' " \
"AND JobTitle != 'FIREFIGHTER'"

# KEEP THIS. It will display the whole dataframe.
df = pd.read_sql(query, conn)
df
```

Out[46]:

	AverageBasePay	AverageBenefits	AverageOvertimePay	SumOfAverages
0	101657.66285	36528.450125	26348.270494	164534.38347

3.1.4: 7 Points

Finally, we'll make our own table in our database.

- Separate the **Salaries table** by **years (2010-2015)**, and add it back to the database.
- Using a loop might be helpful.
- You may use basic python to complete the task. However, using querying on SQL is **mandatory**.
- Feel free to **use multiple lines of code for this problem only**.

Hint: Check [this](#) out

```
In [47]: for year in range(2010, 2016):
        query = f"SELECT * FROM Salaries WHERE Year = {year}"

        dff = pd.read_sql(query, conn)

        table_name = f"Salaries_{year}"
        dff.to_sql(table_name, conn, if_exists='replace', index=False)
```

```
In [48]: # Run this code to check if you successfully added your table.
        cursor = conn.cursor()
        cursor.execute("SELECT name FROM sqlite_master WHERE type='table';")
        print(cursor.fetchall())
```

```
[('Salaries',), ('Salaries_2010',), ('Salaries_2011',), ('Salaries_2012',), ('Salaries_2013',), ('Salaries_2014',), ('Salaries_2015',)]
```



Part 4: Pandas 'Group By'(40 points)

This flowchart is taken from our lecture class presentation and illustrates the process of transforming data using the Pandas GroupBy operation. First, the data is input, followed by applying the GroupBy function to one or more columns of the DataFrame. Once the data is grouped, an aggregation function (such as sum(), mean(), or count()) is applied to compute summary statistics for each group



Your task is to translate the workflow shown in the flowchart into Pandas queries that perform these operations step by step.

Notes: Your task is to translate the workflow shown in the flowchart into Pandas queries. Ensure that the **exact input and exact output** from the flowchart are replicated using Pandas queries, step by step.

4.1 (Point 10) Create a sample dataset that includes columns for account, order, and ext price.

```
In [49]: # Create the Sample dataset from above flowchart
input_data = {
    'account': ['383080', '383080', '383080', '412290', '412290', '412290', '412290',
    'order': ['10001', '10001', '10001', '10005', '10005', '10005', '10005', '10005',
    'ext_price': [235.83, 232.32, 107.97, 2679.36, 286.02, 832.95, 3472.04, 915.12,
}

# Create DataFrame
df = pd.DataFrame(input_data)

# Display Dataframe (DONT REMOVE THE CODE)
print("Input DataFrame:")
df
```

Input DataFrame:

Out[49]:

	account	order	ext_price
0	383080	10001	235.83
1	383080	10001	232.32
2	383080	10001	107.97
3	412290	10005	2679.36
4	412290	10005	286.02
5	412290	10005	832.95
6	412290	10005	3472.04
7	412290	10005	915.12
8	218895	10006	3061.12
9	218895	10006	518.65
10	218895	10006	216.90
11	218895	10006	-72.18

4.2 (Point 10+10 =20) Group by order and show the intermediate results

```
In [50]: # Group by 'order' and show intermediate result
print("Group by 'order':")
groupByOrder = df.groupby('order')

# Display intermediate result for each group; hints: you have to use 'for loop'
for order, group_order in groupByOrder:
    print(f"\nOrder {order}: ")
    print(group_order[['order', 'ext_price']])
```

Group by 'order':

Order 10001:

	order	ext_price
0	10001	235.83
1	10001	232.32
2	10001	107.97

Order 10005:

	order	ext_price
3	10005	2679.36
4	10005	286.02
5	10005	832.95
6	10005	3472.04
7	10005	915.12

Order 10006:

	order	ext_price
8	10006	3061.12
9	10006	518.65
10	10006	216.90
11	10006	-72.18

4.3 (Point 10) Apply the Sum Aggregation for Each Group

Now we'll apply the sum aggregation to get the total ext price for each order:

```
In [51]: # Repeat group by 'order' again and then apply aggregation (sum of 'ext_price' for
print("Sum Aggregation for Each Group: ")

totalByOrders = df.groupby('order')['ext_price'].sum()

# Show the aggregated result after re-groupin (DONT REMOVE THE CODE)
totalByOrders
```

Sum Aggregation for Each Group:

```
Out[51]: order
10001      576.12
10005     8185.49
10006     3724.49
Name: ext_price, dtype: float64
```

4.4 (Point 10) Combine the Results into One Final Table Finally, we will **reset the index** and create a combined table that shows order and the sum of the ext price for each group:

Notes: In pandas, 'reset_index()' is a method used to reset the index of a DataFrame to its default integer-based index. By default, when you perform certain operations like groupby(), the resulting DataFrame may have a new index (e.g., the grouped column). The reset_index() method allows you to convert the current index back to a default sequential integer index and optionally, move the current index values into a regular column.

```
In [52]: # Reset index to combine result into a single DataFrame
finalResult = totalByOrders.reset_index()
# Rename the columns for clarity
```

```
finalResult.columns = ['order', 'Order_Total']  
# Show the final result (DONT REMOVE THE CODE)  
print('Final Result')  
finalResult
```

Final Result

```
Out[52]:
```

	order	Order_Total
0	10001	576.12
1	10005	8185.49
2	10006	3724.49

Bonus(1 point): Assignment Submission with Jupyter Notebook file(ipynb) and PDF

In addition to the instruction of the special bonus question included exclusively in Assignment 1, we would like to highlight the importance of developing good practices for assignment submission in data science projects.

Think of yourself as a reviewer of your peers' work: what would you pay attention to if you had to grade their assignments? Clearly, the raw dataset itself is not the main point—why would someone need to see every line of data you downloaded? Data science is about analysis and insight, not just data collection. What truly matters is how you analyze the data and how you present your results.

For a clean and effective submission, please avoid printing entire datasets unless they are directly relevant to the presentation of your output or if you are asked to do so. Instead, focus on:

- Showing your algorithm with clear, concise code
- Displaying the key results you want us to evaluate (or the outputs you relied on while working)
- Adding a short explanation when asked to interpret your findings

For this bonus question, please submit both your .ipynb file and the PDF version. To create a PDF: go to File (top left corner in Jupyter Notebook) → Print → Destination → Save as PDF.

Are you noticing an overwhelming amount of unnecessary printed results in your PDF? This is exactly why we emphasize the importance of cleaning up your work before submission. Removing trivial data printing makes your notebook look much more professional and easier to review—just like proofreading an essay for good grammar.

Your Homework 1 submission should be lightweight and focused, since this is a starter project designed to help you ease into the semester. :)

This practice helps us grade more efficiently and ensures fairness, since we can always reference your original notebook if regrading is needed.